

UPDATE OF THE ADAPTATION PLAN TO CLIMATE CHANGE HEALTH SECTOR

This Sectoral Adaptation Plan Update has been prepared by the Operational Committee of the Ministry of Health with the technical counterpart of the Ministry of the Environment.

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1. EXECUTIVE SUMMARY

Climate change poses a significant threat to health, affecting clean air, drinking water, food, and housing. It is projected to cause an additional 250,000 deaths annually between 2030 and 2050, with particularly severe impacts on malnutrition, infectious diseases, and heat stress. In Chile, climate effects manifest as extreme events such as heat waves, droughts, floods, and seasonal changes, all of which impact health and require preparedness to mitigate their consequences.

In this context, the current sectoral plan for adaptation to climate change presents 15 measures, including the promotion of sustainable food systems, epidemiological surveillance of climate-related diseases, and strengthening the response to heat waves and outbreaks of vector-borne diseases. These measures aim to integrate approaches to resilience, food security, monitoring, and capacity building within the healthcare network.

To implement this, a comprehensive adaptation strategy is being promoted, including training for healthcare personnel, modernizing health surveillance, and improving governance for effective climate management. Intersectoral coordination and investment in infrastructure and technology are key to improving the response to climate-related emergencies.

The plan is aligned with the guidelines of the Framework Law on Climate Change and Chile's Long-Term Climate Strategy, seeking to reduce vulnerability and increase the resilience of the population to projected climate risks.

2. BACKGROUND

2.1.- Characterization of the sector and its vulnerability

The effects of climate change pose a clear threat to people, their health, and their well-being. This phenomenon will directly impact ecosystem services and the human systems that depend on them. Scientific analyses indicate an increasing trend in the frequency and intensity of extreme weather events over the past fifty years, and it is considered likely that high temperatures, heat waves, and heavy rainfall will continue to become more frequent in the future, which could be disastrous for humanity (MMA, 2014).

The World Health Organization (WHO) has indicated that climate change represents one of the greatest health challenges that the international community will face in the coming decades, estimating that, globally, the effects of climate change on population health will be especially negative, influencing the social and environmental determinants of health, namely, clean air, drinking water, sufficient food and safe housing, and that areas with poor health infrastructure, mostly in developing countries, will be the least able to prepare for and respond to these effects.

According to WHO projections, climate change is expected to cause approximately 250,000 additional deaths each year between 2030 and 2050 due to malnutrition, malaria, diarrhea, and heat stress. Of these, 38,000 are attributed to heat exposure in the elderly, 48,000 to diarrhea, 60,000 to malaria, and 95,000 to childhood malnutrition.

It also estimates that the cost of direct health damage (i.e., excluding costs in sectors that determine health, such as agriculture and water and sanitation) will be between US\$2 billion and US\$4 billion per year by 2030 (WHO, 2024).

Faced with this scenario, it is necessary to develop plans and measures for adapting to climate change, which aim to reduce vulnerabilities and minimize the negative effects of this phenomenon, in order to protect people's health, in accordance with the mission that the Ministry of Health has set for itself in matters of public health, which is: "to lead the health strategies that allow improving the health of the population by exercising the regulatory, normative, surveillance and oversight functions that correspond to the State of Chile, in matters of health promotion, prevention and control of diseases, aligned with the priorities of the government program for the sector and the health objectives of the decade, contributing to sustained progress in the quality of health and care of the inhabitants of the country, ensuring timely access and respect for the dignity of people, with a focus on gender and social inclusion of vulnerable groups, especially people in situations of disability, indigenous peoples and migrants" (SSSP, MISSION. 2024).

This is complemented by the mission of the Undersecretariat of Healthcare Networks, which states that its mission is "to regulate and supervise the operation of healthcare networks through the design of policies, standards, plans, and programs for their coordination and articulation, enabling them to meet the healthcare needs of the user population, within the framework of health objectives, with quality and user satisfaction" (SSRA. MISSION. 2024), thus allowing the incorporation of measures into

Adaptation actions aimed at improving climate resilience levels in the healthcare network. The health sector developed an initial "Health Sector Climate Change Adaptation Plan 2016-2021" early on, recognizing the importance for the sector of implementing significant measures to reduce vulnerabilities and address the effects of climate change. The measures in this first plan were determined taking into account the expected national health impacts of climate change. Its formulation involved the participation of various stakeholders.

from the various departments, units, and offices belonging to the two ministerial undersecretariats that make up the Ministry of Health. One of the fundamental aspects taken into account to address the challenge of climate change and its impact on people's health was to adopt an intersectoral approach to managing the risks and problems associated with the expected changes in climatic phenomena.

The recently approved Framework Law on Climate Change (LMCC) established a State policy on climate management that aims to address the challenges presented by climate change, transition towards development with low emissions of greenhouse gases and other climate forcing factors, until achieving and maintaining greenhouse gas emissions neutrality by 2050, adapt to climate change, reducing vulnerability and increasing resilience to the adverse effects of climate change, and comply with the international commitments assumed by the State of Chile in this matter.

Within the framework of the policies and guidelines established by the National Disaster Prevention and Response Service (SENAPRED), and in accordance with the Framework Law on Climate Change, the Risk and Disaster Management Committees (COGRIDs) are formed to coordinate and implement preventive and response actions in emergency situations arising from climate change. COGRIDs are composed of representatives from various entities and sectors. Collaboration and coordination among COGRID members are essential to ensure an effective and timely response to any eventuality, minimizing the impact on the population and facilitating post-disaster recovery. Furthermore, integrating climate change policies into risk and disaster management strengthens community resilience and contributes to sustainable development.

In this context, the **National Policy and the National Strategic Plan for Disaster Risk Reduction (DRR) 2020-2030** provide a comprehensive framework for planning and implementing actions that address both risk reduction and adaptation to climate change.

As of January 2023, SENAPRED has the power to supervise compliance with all these instruments, ensuring coherent and effective management of risks and disasters in Chile.

Specifically regarding Sectoral Adaptation Plans to Climate Change, the law states that these plans identify, but are not limited to, 12 sectors. They will establish the set of actions and measures necessary to adapt the most vulnerable sectors to climate change and increase their climate resilience, in accordance with the adaptation objectives and targets defined in the Long-Term Climate Strategy. In the case of the health sector, which already has an initial Adaptation Plan, the law stipulates that this plan must be updated within the same timeframes established for the other sectoral plans (Law No. 21,455).

For its part, the Long-Term Climate Strategy (LTCS) provides guidelines for adaptation at the national level, identifying a set of 46 long-term objectives, which reflect similar sectoral response needs, although with their respective specificities, which can be summarized as follows:

- Strengthen sectoral governance at different territorial scales, building capacity in public institutions and among diverse stakeholders in each sector, and deepening citizen participation in the development of sectoral climate change policies.
- To deepen and broaden scientific knowledge, technological development in the various sectors, as well as the interactions and synergies between them, to support decision-making.
- Reduce the risk from the impacts of climate change, through an integrated approach in policies and instruments, that reduce vulnerability, exposure and increase adaptive capacity, monitoring the evolution of risk based on changing climate threats.
- Integrate climate change into sectoral policy instruments at the territorial level, recognizing the diversity of impacts and local realities.
- Implement actions prioritizing Nature-Based Solutions.
- To protect, restore, prevent and reduce the degradation of ecosystems by promoting the sustainable use, production and consumption of natural resources.
- To promote sustainable productive skills in the fisheries and aquaculture, agricultural, livestock and forestry sectors, considering adaptation to climate change in order to contribute to food security and the sustainability of natural resources, and
- Promote water security for human consumption and sanitation, and the provision of water for ecosystems and productive activities, contributing to the efficient use of water resources in the territories. (This guideline is cross-cutting and will be observed in the fulfillment of the other objectives, goals, guidelines and directives of the Strategy)

(ECLP, MinAmbiente. 2021).

Action for Climate Empowerment (ACE) is an internationally recognized instrument for promoting climate action, included in Article 6 of the United Nations Framework Convention on Climate Change (UNFCCC) on education, training, and public awareness. ACE comprises six cross-cutting and interconnected elements considered fundamental to achieving the UNFCCC and Paris Agreement goals of keeping global warming well below 2°C and ideally below 1.5°C, as well as the national objective of achieving carbon neutrality and climate resilience by 2050 at the latest. In this context, Chile has a Climate Capacity Building and Empowerment Strategy, as part of the means of implementation of its Long-Term Climate Strategy (LTCS).

2.2.- Assessment of adverse effects of climate change and projected risks

Climate change has altered marine, terrestrial, and freshwater ecosystems worldwide. The effects are becoming more widespread and have far-reaching consequences.

predicted. It has caused losses of local species, increased diseases and mass mortality events of plants and animals, resulting in the first climate-induced extinctions, the restructuring of ecosystems, increased areas burned by wildfires, and a decline in key ecosystem services. Climate-induced impacts on ecosystems have caused quantifiable economic and livelihood losses and have altered cultural practices and recreational activities worldwide (IPCC_AR6_WGII_TechnicalSummary).

Climate change is already putting pressure on food and forest systems, with negative consequences for the livelihoods, food security, and nutrition of hundreds of millions of people, especially in low and mid-latitudes. The global food system is failing to address food insecurity and malnutrition in an environmentally sustainable manner (IPCC_AR6_WGII_TechnicalSummary).

Currently, approximately half of the world's population experiences severe water scarcity for at least one month per year due to climatic and other factors. Water insecurity manifests itself through water scarcity and climate-induced hazards and is further exacerbated by inadequate water governance. Extreme events and underlying vulnerabilities have intensified the social impacts of droughts and floods, negatively impacted agriculture and energy production, and increased the incidence of waterborne diseases. The economic and social impacts of water insecurity are more pronounced in low-income countries than in middle- and high-income countries (IPCC_AR6_WGII_TechnicalSummary).

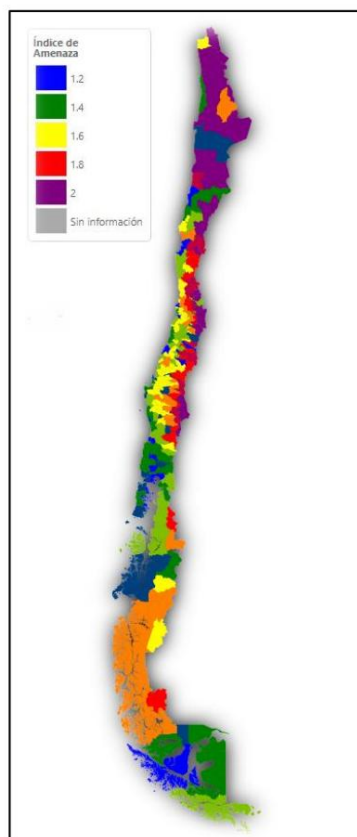
Climate change has already harmed people's physical and mental health. In all regions, the health impacts often undermine efforts to achieve inclusive development. Women, children, the elderly, indigenous peoples, low-income households, and socially marginalized groups within cities, settlements, regions, and countries are the most vulnerable (IPCC_AR6_WGII_TechnicalSummary).

Since the IPCC Fifth Assessment Report, there is growing evidence that climate hazards associated with extreme events and variability act as direct drivers of involuntary migration and displacement, and as indirect drivers through the deterioration of climate-sensitive livelihoods. Most climate-related displacement and migration occurs within national borders, and international movements occur mainly between countries with contiguous borders. Since 2008, an annual average of more than 20 million people have been internally displaced by climate-related extreme events, the most common of which are climate-related extreme events (IPCC_AR6_WGII_TechnicalSummary).

being storms and floods

Climate change will increase the number of deaths and the global burden of infectious and non-communicable diseases. More than nine million climate-related deaths are projected annually by the end of the century in a high-emissions scenario that takes into account population growth, economic development, and adaptation. Health risks will differ according to gender, age, region, and income (IPCC_AR6_WGII_TechnicalSummary).

status social and



Change in the annual average of the daily maximum temperature during the period 2035-2064 (projected considering the RCP8.5 scenario) with respect to the average of the maximum temperature observed during the period 1980-2010.

Source: arclim.mma.gob.cl/atlas/view/aumento_mortalidad_calor_addcbit



It represents the variation in the incidence of heat waves between the historical climate (1980-2010) and the future climate (2035-2065 under the RCP8.5 scenario). All variations are positive (increase in heat waves) but with different changes across the country.

Source: arclim.mma.gob.cl/atlas/view/extreme_heat_human_health

As noted in the preceding paragraphs, exposure to the various anticipated effects of climate change could impact the health of millions of people globally, and Chile is no exception. Our country is particularly vulnerable to the manifestations and effects of climate change due to its complex geography, as it is subject to numerous variations in climatic conditions and extreme hydrometeorological events. This underscores the need to not only reduce exposure but also prepare to respond to climate change contingencies and strengthen social and institutional capacities at the national level.

According to what was indicated in the first PACC-S, it is possible to differentiate three routes by which climate change affects health:

- 1) Direct impacts, which are mainly related to changes in the frequency of extreme events including heat waves, floods and storms.
- 2) Effects mediated by natural systems, for example, disease-carrying vectors, outbreaks of waterborne diseases, and air pollution.

3) Effects strongly mediated by human systems, for example, occupational impacts, malnutrition and mental stress (IPCC, 2014).

For the purpose of organizing the analysis and visualizing the relationships between the direct and indirect manifestations of climate change and its effects, it is useful to group them into three large groups or forms of manifestation:

- Trends, understood as those changes whose occurrence is observed in a sustained manner over the medium or long term, regardless of minor variations that may occur from year to year. Examples include the increase or decrease in rainfall in a given geographic area, the general acidification of the oceans, variations in average temperatures in certain zones or locations, etc.
- Extreme weather events, understood as those events of sudden or gradual occurrence associated with climatic phenomena that manifest with particular intensity and/or persistence, such as, among others, droughts that are prolonged and intensified in certain areas of the planet, intense rainfall that is the origin of mudslides and floods, and cold or heat waves.¹ • Changes in the seasonality of events, which correspond to isolated events, often of short duration, but intense, that occur in times or seasons of the year in which they did not usually occur, such as unusually warm days that occur in the cold season or unusual rainfall in the dry season.

Among the direct and indirect climate threats associated with extreme weather events that have occurred in our country in recent decades, with a significant impact on people's health and safety, are mudslides, floods, drought, storms, cold waves, heat waves, landslides, coastal flooding, landslides, storm surges, tornadoes, waterspouts, snowstorms, and sandstorms. It is important to note that some of these threats have occurred with greater frequency and/or intensity than usual, such as waterspouts, tornadoes (2019, Maule, Ñuble, Biobío, and Araucanía regions), and forest fires (2017 and 2023, south-central zone). urban and peri-urban fires (Valparaíso, 2014; Viña del Mar, 2022; Valparaíso Region, O'Higgins, Maule, Biobío, Araucanía, Los Lagos, 2024) or even,

Some threats, such as sandstorms (Arica, 2016; Diego de Almagro, 2022), have little historical precedent in certain areas of our country. It should be noted that specific risk analyses or health impact assessments have not yet been conducted in declared latent zones.

The main health risks associated with the various manifestations of climate change identified in the first PACC-S and the associated climatic phenomena are shown in the following table:

¹ According to the IPCC, the increase in extreme weather and climate events has led to some irreversible impacts as natural and human systems are pushed beyond their capacity to adapt. IPCC, 2022: Summary for Policymakers.

Table 1.- Health impacts and their association with climatic phenomena

Expected health impacts: Increase	Associated climatic phenomena
in infectious diseases, diarrheal diseases and malnutrition.	Droughts and extreme rainfall (the latter being a factor in floods, river overflows, landslides): are expected to cause a decrease in the quality and quantity of available water and food.
Variation in the distribution and frequency of diseases transmitted by mosquitoes, ticks and other vectors.	
Increase in injuries and deaths.	
Increase in psychological disorders ¹ .	Increased humidity or drought conditions ² . These events are associated with the proliferation of infectious vectors that transmit diseases mentioned.
Increase in cardiorespiratory and allergic diseases	Droughts: as a consequence, they would generate persistent pollution episodes due to the decrease in rainfall. Increase in average temperature, tropospheric ozone concentrations, and airborne allergens. Increased extreme temperatures and increased likelihood of acute pollution episodes caused by forest fires.
Health crisis in vulnerable people (children, the elderly, pregnant women, low-income populations, people periodically exposed to the outdoors and those with pre-existing respiratory conditions) people, with diseases primarily diseases and cardiovascular).	Increase in extreme temperatures.
Increase in <i>Vibrio</i> - associated diseases <i>parahaemolyticus</i> and harmful algal blooms.	Increase in sea and inland water temperatures.
1. The effects on mental health can be acute and direct, indirect, and psychosocial. The former are associated with the occurrence of extreme events that trigger psychological trauma linked to somatic illnesses, depression, and drug and alcohol abuse. Indirect impacts, on the other hand, generate subsequent reactions such as anxiety, grief, anguish, and numbness, among others. Finally, psychosocial impacts affect mental health due to their effects at the community level. These are generated by different climatic impacts, for example, heat waves that can create inter-group conflicts due to violent reactions from the population, or drought; by economic impacts that result in resource scarcity; and by migrations associated with the phenomenon (Garibay M. & Bifani-Richard P. 2012). 2. Climate projections for the northern region differ in their results, with some indicating an increase in precipitation (and a consequent increase in humidity) and others a decrease (and a consequent increase in the probability of droughts). However, in both cases, there would be an increase in the proliferation of disease vectors and the corresponding health impacts mentioned. It is noted that drought conditions can increase vector-borne diseases due to their connection with the supply of drinking water and/or irrigation through channels other than those currently used, which could generate uncontrolled vector reservoirs, especially for mosquitoes that transmit dengue fever.	

Adapted from Tables 1 and 2 of the PACC-S 2016-2021

Finally, this plan aims to ensure that at least its actions are included in accordance with the Plan National Adaptation to Climate Change (PNACC) specifically with Specific Objective No. 2:

“Implement cross-cutting actions to reduce risk and vulnerability to climate change and strengthen resilience,” and with Specific Objective No. 3, “Develop the means of implementation for resilience at the national, regional, and community levels,” by building capacity within the health system and reducing the sector’s vulnerability to extreme events through preventive actions, reducing vulnerability to infectious diseases transmitted by vectors, contaminated water or food, and transmission among people of lower socioeconomic strata. Finally, by promoting and informing the population about the impacts of climate change on health in order to encourage the adoption of self-care measures and community organization to reduce vulnerability.

In the context of the PACC-S (2016-2021), the measures included were either implemented or reformulated in the current plan, with the exception of Measure No. 7: “Development of disease burden associated with climate change” and Measure No. 8: “Development of a study projecting demand in the hospital network due to climate change and associated costs,” which were not carried out within the stipulated timeframe. This is primarily because epidemiological surveillance of climate change and its effects on health is conducted, identifying tracer diseases, both communicable and non-communicable, associated with climate change and their impact on hospital discharges, mortality, and emergency room visits. A disease burden study typically requires planning a year in advance, significant financial resources, and approval from the relevant authorities, and must be prioritized by the Health Authority. Therefore, given its high cost and very low benefit, this proposal was reformulated into other measures presented in the new plan. Measure No. 8 is a measure that can be implemented if measure No. 7 is fulfilled, so they are directly related and the non-implementation of the first implies the non-implementation of the later one.

3. OBJECTIVES AND MEASURES

1.1.Objective

Increase the current and future resilience of the health sector to respond to climate change, focusing especially on the most vulnerable groups of the population, while ensuring the proper functioning of health services and networks, considering the new climatic conditions and projected impacts.

1.2. Specific Objectives

1. Promote healthy and sustainable eating, and reduce the risks of foodborne illnesses caused by climate change.
2. Reduce the risks of vector-borne diseases caused or increased by climate change.
3. Strengthen knowledge and dissemination about the impacts of climate change in health.

4. Reduce the risks of extreme events to human health.
5. Strengthen water security to guarantee access to safe and clean water.

1.3. Presentation of Measures

Strategic lines are sets of strategic objectives or vertical combinations thereof. They are based on broad strategic concepts that aim to focus an organization, significantly guiding all actions within strategic planning. Within these strategic lines (SL), we find the following for the healthcare sector:

- E1. Health education, training, awareness and cooperation.
- E2. Analysis and research on impacts and scenarios.
- E3. Intersectoral and interinstitutional coordination.
- E4. Climate risk reduction.
- E5. Vulnerability assessment and preparedness for climate events.

Thus, the measures formulated according to the strategic lines are formulated as follows:

Specific Objectives (SO)	Strategic lines (SL)	Measures
Objective 1: Promote healthy and sustainable eating, and reduce the risks of climate change-related diseases for food.	E1. Health education, training, awareness and cooperation.	Measure No. 1: Promotion of nutrition healthy and sustainable.
	E2. Analysis and research on impacts and scenarios.	Measure No. 2: Analysis of the dynamics between the Climate change and foodborne illnesses.
	E3. Intersectoral and interinstitutional coordination	Measure No. 3: Prevention and management of poisonings caused by harmful algal bloom (HAB) events in marine waters.
Objective 2: Reduce the risks of vector-borne diseases caused or exacerbated by climate change.	E2. Analysis and research on impacts and scenarios.	Measure No. 4: Monitoring and communication system for unusual findings, in time and space, of vectors and reservoirs of public health importance prioritized* that are associated with climate change.
	E4. Climate risk reduction.	Measure No. 5: Strengthening the response capacity of the healthcare network to vectors and reservoirs.
Objective 3: Strengthen knowledge dissemination on the impacts of climate change on health.	E1. Health education, training, awareness and cooperation.	Measure No. 6: Strengthening of public health human capital.
	E4. Climate risk reduction.	Measure No. 7: Epidemiological surveillance system for health effects associated with climate change.
Objective 4: Reduce the health risks of extreme events Human.	E4. Climate risk reduction.	Measure #8: Increase in the quantity of services that can be performed online.
	E4. Climate risk reduction.	Measure No. 9: Development of a response plan for the healthcare network to heat waves.

	E4. Climate risk reduction.	Measure No. 10: Apply a methodology that allows the development of regional emergency plans considering climate threats along with other local threats.
	E5. Survey of vulnerabilities and preparedness for climate events.	Measure No. 11: Diagnosis and implementation of Vital Lines and Climate Threats in Health Facilities.
	E4. Climate risk reduction.	Measure No. 12: Development of an internal plan of the Ministry of Health for the response to extreme high temperature events and heat waves.
Goal 5: Strengthen water security to ensure access to safe and clean water	E2. Analysis and research on impacts and scenarios.	Measure No. 13: Sanitary control of contaminants emerging in freshwater bodies.
	E2. Analysis and research on impacts and scenarios.	Measure No. 14: Conduct a vulnerability assessment on territorial availability of water resources and sanitary conditions for their consumption (other organizations).
	E2. Analysis and research on impacts and scenarios.	Measure No. 15: Modernize the sanitary surveillance of water for human consumption through digital applications with a risk-based approach.

4. ADAPTATION MEASURES

4.1.- Adaptation Measure No. 1: Promotion of healthy eating and sustainable.

Climate change that has occurred on the Earth's surface since pre-industrial times, and which has doubled the air temperature, is a situation that negatively affects food security and terrestrial ecosystems, which is why sustainable management is necessary throughout the food chain, from food production to consumption (WMO UNEP, 2019).

Regarding food consumption and messages aimed at encouraging healthy eating choices, the latest recommendations for updating Food-Based Dietary Guidelines (FBDGs) suggest incorporating environmental sustainability and paying greater attention to the sociocultural factors of different population groups and the rapid changes in dietary trends (Herforth A, 2019). Along these lines, the Food and Agriculture Organization of the United Nations (FAO) recommends that governments commit to incorporating sustainability into the development and dissemination of FBDGs (FAO, 2018). Thus, the Dietary Guidelines for Chile, in effect since 2022, consider a healthy diet to be the consumption of varied and balanced foods that meet people's physiological and nutritional needs throughout their lives. These foods should be fresh, natural, and minimally processed, promoting physical, mental, and social well-being. In this sense, sustainable diets not only promote human health, but also minimize environmental impact, contributing to the preservation of biodiversity and the health of the planet.

Sustainable diets generally have a low environmental impact for the following reasons: they promote diversity, which implies the consumption of a wide variety of foods; they are based on minimally processed foods, especially those grown in the field; they promote moderate consumption of meat and dairy products; very limited consumption of foods high in fat, sugar or salt and low in micronutrients, for example, french fries, confectionery products and sugary drinks; and consumption of drinking water instead of soft drinks and sodas (FAO, 2018).

Sustainable food systems align with sustainable development by using natural resources efficiently, promoting responsible agricultural practices, and supporting local agriculture, thereby reducing the carbon footprint. By prioritizing local and seasonal foods, the use of fossil fuels needed to transport food products over long distances is reduced, which is essential for lowering greenhouse gas emissions. Sustainable water use is another fundamental pillar of sustainable food systems. Plant-based food production generally requires less water compared to animal-based foods. An illustrative example is the water footprint: producing a tomato requires 13 liters of water, while a hamburger can require up to 2,400 liters. According to FAO data, current food production is damaging the environment, generating between 20% and 30% of greenhouse gas emissions, leading to land-use change and biodiversity loss. In general, the food production systems included in sustainable food guidelines are those whose production and distribution methods have prioritized socially and environmentally sustainable systems (FAO, 2018).

Promoting healthy and sustainable eating, as recommended in the Dietary Guidelines for Chile, is not only crucial for improving the health of the population, but also for reducing environmental impact, fostering more resilient food systems in the face of climate change and guaranteeing food security for future generations.

Measure No. 1: "Promotion of healthy and sustainable eating"			
Element	Sub-element	Content	
Measurement	Aim Specific (OE)	Promoting healthy and sustainable food in the context of climate change	
	Lineament Strategic (LE)	E1 Health education, training, awareness and cooperation.	
	Description and expected result of the measure	Promote healthy and sustainable eating practices that prepare the population for the challenges climate change will bring to food security. Some healthy and sustainable eating systems that could be considered include the Planet Diet Model, the Lancet EAT Commission, or FAO guidelines. GOAL: Defined healthy and sustainable dietary recommendations.	
	Indicator Measurement	1. (Number of activities completed / number of activities planned) Number of activities carried out = dissemination activities carried out annually in the context of the implementation of the Dietary Guidelines, within the framework of the Climate Change Adaptation Plan.	
	Institutions	Responsible	Department of Nutrition and Food of the Division of Healthy Public Policies and Promotion (DIPOL).
	Reach	Beneficiaries	National population accessing primary healthcare services. Health benefits for the population through healthier and more sustainable food.
		Territorial	national
Synergies	It incorporates or contributes to the cross-cutting guidelines of ECLP, PNACC, NDC	It relates to the following ECLP objectives (health sector): Objective 1: To strengthen knowledge and evidence at the national level regarding environmental variables linked to climate change and how these could affect people's health and well-being. Objective 2: To educate, empower and involve citizens in the issues of climate change and its impacts on the health and well-being of the population, both nationally and regionally, to raise awareness in communities and promote prevention, preparation and response measures to the expected impacts of climate change on health.	

	Synergy with other public policies	- Synergy with the SDGs: 2 Zero Hunger, 3 Good Health and Well-being, 13 Climate Action. - Active participation: providing the community with the tools and knowledge to adopt sustainable diets. - It contributes to the PSM mitigation measures, associated with the reduction of greenhouse gases, derived from food waste. Gender-responsive NOTE: The updated Dietary Guidelines for Chile address healthy eating from a gender perspective, recognizing that women currently still have a double workload, with work outside the home and care work for their family members. Therefore, specific recommendations are provided on sharing caregiving tasks within the family and extended family.													
Planning	Timeline Implementation (Main activities that must be carried out to achieve the goal of the (Measurement)	Actions (Specify the name of the shares)	Years					Indicators Actions (quantitative or qualitative)					Institutions		Total cost by action
			Year 1 2025	Year 2 2026	Year 3 2027	Year 4 2028	Year 5 2029	Name	Meta	Unit of measure	Medium of Verification	Categorization Media of Implementation (T, C, F u OJ2)	Responsible	Contributor ^a	
		Action 1: Disseminate, including the Primary Care and Intersectoral, on the practices of Healthy Eating through the implementation of Dietary Guidelines for Chile, with special emphasis on the dimension environmental of the Nutrition, and the Change approach Climate, incorporating the sustainability model to promote and strengthen healthy diets based on systems food sustainable, promoting co-responsibility in families and communities, reducing environmental impact and contributing to food and nutritional security current and future generations.	x	x	x	x	x	Spread the word Food ion Healthy +	Spread the word Food ----- Healthy +	Number of activity dissemination activities carried out annually the Document action Official	Annual report of activities of diffusion carried out in the context of the implementation of the Guidelines Food, in the framework of the Plan Adaptation to Change Climatic.	Creation and strengthening to of capabilities	Department the of Nutrition and Food Division Policies Public Health and Promotion (DIPOL).	-SEREMIS of Health - Attention Primary Health - Programmatic offering of other Divisions of Ministry of Health - MINAGRI - MINECON MDSF (Choose Healthy Living). MMA3 Ministry of Science, Vicerrectory of Education, (JUNJI- JUNAEB); MINAGRI (ODEPA), MMEG, SUBPESCA	M\$669.633
		Cost	M\$16 6.227	M\$10 4.727	M\$16 6.227	M\$66. 227	M\$16 6.227								
		Total cost per year	M\$16 6.227	M\$10 4.727	M\$16 6.227	M\$66. 227	M\$16 6.227								
Financing	Estimated cost of the measure (M\$)	M\$669.633													
	Possible sources financing	To be defined by the Ministry of Health. Additional budget is required.													
	Assumptions	- Sufficient funding exists for the development of the proposed activities - There is availability of time on the part of health care officials to access training on dietary guidelines - That the officials of the nutrition programs of the SEREMIS of Health of the country promote the implementation of the GABAS.													

4.2.- Adaptation Measure No. 2: Analysis of the dynamics between climate change and foodborne diseases

Foodborne illnesses represent a significant global burden of disease due to their high morbidity and mortality rates, particularly among vulnerable groups such as children under five, pregnant women, the elderly, and immunocompromised individuals. According to WHO estimates, 600 million cases are reported annually.

² Indicate if the action can be categorized within any of the following means:

T: Technology development and transfer.

C: Capacity building and strengthening

F: Financial guidelines.

O: Other.

³ It is expected that the MMA will be able to collaborate with technical support for the development and dissemination of best practices in feeding Healthy through the implementation of the current Dietary Guidelines for Chile, in the areas of its competence.

and 420,000 deaths associated with foodborne illnesses, with more than 200 types of foodborne diseases identified. These illnesses most frequently present with gastrointestinal symptoms, although cutaneous, neurological, and immunological manifestations can also occur, even leading to systemic manifestations, cancer, and death. In Chile and worldwide, the risk of foodborne illness outbreaks has become an emerging problem due to factors such as the globalization of food trade, technological changes in the food industry, changes in eating habits and migration, and the emergence or re-emergence of etiological agents. Undoubtedly, climatic variables such as temperature and humidity can affect these factors, increasing the risk either by creating conditions that increase the frequency and severity of these illnesses or by amplifying their impact. Based on this background, this new adaptation measure, incorporated into this updated sectoral plan, aims to compare the behavior of foodborne illness outbreak variables with climate factors.

Measure No. 2: “Analysis of the dynamics between climate change and foodborne illnesses”															
Element	Sub-element	Content													
Measurement	Aim Specific (OE)	To contribute to understanding the dynamics between climate change and foodborne illnesses, with geographical relevance													
	Lineament Strategic (LE)	E2 Analysis and Research on Impacts and Scenarios													
	Description and expected result of the measure	Review the environmental epidemiological surveillance of foodborne disease outbreaks (FBD) considering 2 stages: 1.- Select and compare the behavior of variables of foodborne disease outbreaks and climatic variables, with a temporal and geographical focus in the period 2012 - 2022, which allows the identification of possible relationships. 2.- Update and reinforce the aforementioned Circular, with a focus on the registration of variables contributing to the proliferation of pathogenic microorganisms in food, particularly those related to the temperature factor. GOAL: To have an updated Circular on Epidemiological Investigation and Environmental Control of Foodborne Disease Outbreaks, which contributes to improving the overall process of surveillance and investigation of these events, increasing the registration of specific diagnoses, and strengthening the identification of contributing factors, especially those related to temperature.													
	Indicator Measurement	1. Report number for review of historical data (2012-2022) on foodborne disease outbreaks and climatic variables. 2. Issuance of an Updated Circular for the investigation of foodborne illness outbreaks.													
	Institutions	Responsible	Department of Nutrition and Food of the Division of Healthy Public Policies and Promotion (DIPOL).												
	Reach	Beneficiaries	The result of the measure benefits the various actors linked to food production to define and establish improvements in production and marketing processes, as well as labeling, and at the same time benefits the health authority in relation to risk management through actions of its own. Scope: Deepening knowledge about the dynamics between climate change and foodborne illnesses												
		Territorial	The measure will be implemented nationwide, taking into account geographical information related to the areas of action (health jurisdiction) of the Regional Health Authorities (SEREMI de Salud).												
Synergies	It incorporates or contributes to the cross-cutting guidelines of ECLP, PNACC, NDC	ECLP: Goal 1.1: By 2025, have relevant indicators to monitor in their respective monitoring zones considering local environmental conditions linked to climate change													
	Synergy with other public policies	- National Health Strategy, Strategic Objective No. 7 “Environment” / Impact Objective 7.2 “Reduce the population exposed to unsafe food”. This measure can contribute to achieving the impact objective “Reduce the population exposed to unsafe food” defined in the ENS 2021-2030. This measure is gender-blind.													
Planning	Timeline Implementation (Main activities that must be carried out for the measure)	Actions (Specify the name of the shares)	Years					Indicators Actions (quantitative or qualitative)					Institutions		Total cost per action
			Year 1 2025	Year 2 2026	Year 3 2027	Year 4 2028	Year 5 2029	Number re	Meta	Unit of measure	Medium of Verification	Categorization Media of Implementation (T, C, F u OI)	Answers	Collaborate time	
		Action 1: Select and compare the behavior of...		x				Inform ...	Inform it is from review	Number of Document	Report of the review of data...	Other	far ento de Nutrition	far ento de	M\$35.912

⁴ Indicate if the action can be categorized within any of the following means:

T: Technology development and transfer.

C: Capacity building and strengthening

F: Financial guidelines.

O: Other.

	achievement of the goal of (Measurement)	variables of foodborne disease outbreaks and climatic variables, with a temporal and geographical focus in the period 2012 - 2022, which will allow the identification of possible relationships						n of data history cos	you officers	historical accounts of outbreaks of illness transmitted due to food and climatic variables		-- Food	Epidemiology MMA5 Academia, Address Meteorologic of Chile, Service Agricultural and Rancher (SAG) ⁵ , SERNAPES	
		Cost	M\$	M\$35.912	M\$	M\$	M\$							
		Action 2: Update and strengthen the The aforementioned circular, with a focus on recording variables that contribute to the proliferation of pathogenic microorganisms in food, particularly those related to the factor temperature	x	x	x	x	x	flow r	flow r	Number of Document you officers	Circular updated	Other	far ento de Epidemiology	far ento de Nutrition -- Food
		Cost	M\$134.74 ⁵	M\$86.500 M\$86.500 M\$86.500	M\$86.500	M\$86.500								
		Total cost per year	M\$134.74 ⁵	M\$122.41 ²	M\$86.500 M\$86.500	M\$86.500								M\$480.745
Financing	Estimate of cost of the measure (M\$)	M\$516.657												
	Possible sources of funding	To be defined by the Ministry of Health. Additional budget is required.												
additional information	Assumptions	- Sufficient funding exists for the development of the proposed activities - The information exists and is available with all the variables required to make the proposed comparisons.												

4.3.- Adaptation Measure No. 3: Prevention and management of poisonings caused by harmful algal bloom (HAB) events in marine waters

"Red Tide" is the term commonly used to refer to Harmful Algal Blooms (HABs), which are dynamic proliferations of phytoplankton microalgae that produce toxins of varying types that affect humans and animals. These compounds accumulate in shellfish, primarily bivalve mollusks, which feed on phytoplankton, without causing them harm when they have lived alongside it for extended periods.

The toxins accumulated in shellfish do not cause organoleptic alterations that suggest contamination, and they share the characteristic of being heat-stable and not altered by the addition of lemon or vinegar.

Marine toxin-producing harmful algal blooms (HABs) have primarily affected southern Chile for the past five decades. However, in the last eight years, they have appeared with increasing frequency, geographic extent, and toxicity in the northern and central regions of the country.

Studies conducted at the international and national levels indicate that Chile is highly vulnerable to

⁵ The MMA is expected, within its area of competence, to contribute as a facilitator in the coordination with other institutions that may support the achievement of the objective of the measure, where relevant.

6. Collaboration through the delivery of information in the following areas

Results of the sanitary inspection (post-mortem inspection).

Unfavorable results in the official microbiological verification and in sampling of the residue control program in slaughterhouses intended for export.

Climate change and climate projections for the country show as main effects an increase in ambient and sea surface temperature and a decrease in precipitation, especially in a large area of the central-southern region, conditions that favor the occurrence of harmful algal blooms (HABs).

Currently, almost all marine toxins described in the literature are recognized in Chile, the most dangerous being the amnesic toxin, the diarrhetic toxin and the paralytic toxin; the latter being the most serious for human health, as it can cause death by respiratory paralysis between 2 and 12 hours after ingestion of contaminated shellfish.

In Chile, 80% of the population consumes seafood (ENCA, 2014), exposing 15,968,711 people to marine toxin poisoning (INE, 2023) with consequences such as gastrointestinal complications, amnesia, paralysis and death.

Considering the lack of antidotes for the treatment of marine toxin poisoning, the lack of technologies to control blooms in the environment or their elimination from shellfish, the disturbances they cause in social and economic systems, and the lack of models that allow predicting their appearance, duration, and location, it becomes imperative to implement public policies that strengthen the prevention of poisoning from the consumption of seafood contaminated with marine toxins in the exposed population.

Measure No. 3: "Prevention and management of poisonings caused by harmful algal bloom (HAB) events in marine waters."			
Element	Sub-element	Content	
Measure	Aim Specific (OE)	Preventing and managing poisonings caused by harmful algal bloom events in marine waters	
	Lineament Strategic (LE)	E3 Intersectoral and interinstitutional coordination	
	Description and expected result of the measure	Generate and strengthen agreements with competent institutions to anticipate and manage harmful algal bloom events with an impact on public health in marine waters, through the updating of early warning systems and the development and implementation of monitoring, surveillance and inspection plans for extraction areas prone to HAB events, in order to act quickly in the face of the occurrence of events in order to prohibit the extraction of seafood products contaminated with marine toxins that cause poisoning and that influence the way of life and income of families involved in the marketing chain of seafood products and other activities associated with the marine environment. GOAL: To contribute to the generation of early warning systems and monitoring plans to prevent the transmission of diseases from contaminated seafood.	
	Indicator Measurement	1. Number of coordination meetings held and documented, with minutes recorded per period (annual)/ number of coordination meetings planned per year. 2. Number of early warning protocols updated/Number of early warning protocols committed to in this Plan. 3. Number of monitoring plans implemented in prioritized areas/Number of monitoring plans committed to in this Plan in areas prioritized.	
	Institutions	Responsible	Department of Nutrition and Food of the Division of Healthy Public Policies and Promotion (DIPOL)
	Reach	Beneficiaries	The program's target population represents 80% of the country's population, specifically those identified as seafood consumers by the National Food Consumption Survey (ENCA, 2014, the latest available study). The intervention effectively reaches all seafood consumers nationwide, as it monitors and oversees all seafood production destined for consumption and marketed through formal channels authorized by the health authority. The number of beneficiaries is projected to reach 16,165,562 by 2025, 16,258,246 by 2026, and 16,347,033 by 2027 (Projection prepared by the Ministry of Health based on the INE's projected population equivalent for 2022 and the proportion of seafood consumers according to ENCA 2014). Reduction in cases of diseases transmitted by seafood contaminated with marine toxins, from formal marketing channels.
		Territorial	The scope of the measure is national.

Synergies	Incorporates or contributes to ECLP's cross-cutting guidelines, PNACC, NDC	ECLP: Goal 1.1: By 2025, have relevant indicators to monitor in their respective monitoring zones considering local environmental conditions linked to climate change. Consideration of types of knowledge: include knowledge from fishing trade associations (linked to criteria of the social pillar of the NDC).													
	Synergy with other public policies	- Synergy with the SDGs: 3 on Good Health and Well-being, 13 on Climate Action, 14 on Life Below Water. - Gender mainstreaming is not relevant													
Planning	Timeline Implementation (Main activities that must be carried out to achieve the goal of the Measure)	Actions (Specify the name of the shares)	Years					Indicators Actions (quantitative or qualitative)					Institutions		Total cost per action
		Year 1 2025	Year 2 2026	Year 3 2027	Year 4 2028	Year 5 2029	Number re	Meta	Unit of measurement	Medium of Verification	Categorization Media Implementation (T, C, F u O)7	Responsible	Collaborators		
		Action 1 Generate agreements with institutions with competence for the prevention and management of poisonings caused by events of harmful algal blooms in marine waters.	x					Agreement of the with the institutions with competence	Of the reunions al year	Number of Document you	Two Minutes of meetings with agreements.	Capacity building and strengthening	Department of Nutrition and Food (DIPOL)	SERNAPESCASUB FISHING IFOP Associations Trade unions of Artisanal Fishing Chilean Navy. DMC SENAPRED	M\$ \$1.712
		Cost	M\$ \$1.712	M\$ 0 M\$ 0	M\$ 0 M\$ 0										
		Action 2: To coordinate the generation and/ or updating of early warning protocols for flowering events of harmful algae in marine waters to act quickly in response to occurrence of these events.		x	Porcen protocol suit caves of alert temp current year zados and valida two, breakdown sado by type of flowering from the algae in marine waters B5.			Protocolos de alert temp cave actual hoists	And protocols current zado	Number of Document axis 5	An updated Early Warning Protocol, according to type of bloom algal one marine waters.	Creation and strengthening of capabilities	Department of Nutrition and Food (DIPOL)	SERNAPESCA SUBPESCA IFOP Associations Trade unions of Artisanal Fishing Chilean Navy.	M\$ 2.000
		Cost	M\$ 0	M\$ 2.000	M\$ 0 M\$ 0 M\$ 0										
		Action 3: Develop and implement annual monitoring, surveillance and auditing plans extraction areas in priority areas and the products of sea once extracted (shellfish).		x	x	x	x	Planes annual It's from prompt oreo	Four planes of warning superior from 2026 al 2029	Number of Document us	- Annual plan monitoring implementad	Creation and strengthening of capabilities	Department of Nutrition and Food (DIPOL)	SERNAPESCA SUBPESCA IFOP Associations Trade unions of Artisanal Fishing Chilean Navy.	M\$3.979.004
		Cost	M\$68.064	M\$97 7.740	M\$97 7.740	M\$97 7.740	M\$97 7.740								
		Action 4 To guarantee the supply of biological reagents and the production of the supplies necessary for the development of surveillance.	x	x	x	x	x	Coordinates was born n	Coordinates nation between sometimes The - ISP - TO BE IS the Health to project takes input biological cos for the year	Number of Document you	Document containing the projection annual supplies biological	Capacity building and strengthening	Department of Nutrition and Food (DIPOL)	Institute Public health SEREMIS of Country's health	M\$19.200

⁷ Indicate if the action can be categorized within any of the following means:

T: Technology development and transfer.

C: Capacity building and strengthening

F: Financial guidelines.

O: Other.

		Cost	M\$ 3.400	M\$3.4 00	M\$ 3.950	M\$ 3.950	M\$ 4.500							
		Total cost per year	M\$76. 615	M\$98 3.140	M\$98 1.690	M\$98 1.690	M\$98 2.240							M\$ 4.005.375
Financing	Estimate of cost of the measure	M\$ 4.005.375												
	Possible sources of financing	To be defined by the Ministry of Health. Additional budget is required												
	Assumptions	- The first assumption is that the funding will be available to carry out the committed activities. - Weather conditions permit the execution of monitoring, surveillance and inspection activities. - There should be no large-scale, geographically extensive, and toxic algal blooms. - There are no threats of natural or anthropogenic origin of great magnitude and territorial coverage. - That the qualified human resources of the regional red tide programs be maintained in the SEREMIS of Health.												

4.4.- Adaptation Measure No. 4: Monitoring and communication system for unusual findings, in time and space, of vectors and reservoirs of public health importance prioritized* that are associated with climate change

Within the framework of the Climate Change Adaptation Plan, Adaptation Measure No. 4 is presented, which aims to create a Monitoring and Communication System for Unusual Findings of Priority Vectors and Reservoirs of Public Health Importance. This new measure seeks to strengthen prevention in mainland Chile, considering that for decades there have been no autochthonous cases of mosquito-borne diseases, vector-borne transmission of Chagas disease (by *Triatoma infestans*) has been interrupted since 1999 (recertified in 2016), and the country declared itself free of canine rabies in 2010. Through the early detection of vectors or reservoirs in territories where their presence has not been confirmed, the aim is to act promptly, preventing potential disease transmission.

Measure No. 4: "Monitoring and communication system for unusual findings, in time and space, of vectors and reservoirs of public health importance prioritized* that are associated with climate change."			
Element	Sub-element	Content	
Measurement	Aim	Reduce vulnerability to the spread of vectors and reservoirs.	
	Specific (OE)		
	Lineament	E2 Analysis and Research on impacts and scenarios.	
	Strategic (LE)	Identification of unusual findings of vectors and reservoirs of public health importance prioritized, in territories where there are no current records of their presence.	
	Description and expected result of the measure	The monitoring system will serve to generate early warnings that allow for the corresponding environmental control actions to be carried out in order to prevent possible cases or outbreaks of the diseases that these vectors or reservoirs transmit. *Those vectors and reservoirs that have implemented surveillance programs will be considered prioritized, such as the mosquito vector program (<i>Aedes aegypti</i> and <i>Anopheles pseudopunctipennis</i>), the vector surveillance program for Chagas disease, and Rabies. GOAL: Implement a Monitoring and Communication System to strengthen the detection and control of unusual events of findings of vectors and animal reservoirs associated with zoonosis programs and vector diseases related to climate change (arboviruses, malaria, wildlife rabies and Chagas disease). Implement a Monitoring and Communication System to strengthen the detection and control of unusual events of findings of vectors and animal reservoirs associated with zoonosis programs and vector diseases related to climate change (arboviruses, malaria, wildlife rabies and Chagas disease).	
	Indicator Measurement	1. Implemented Monitoring Platform	
	Institutions	Responsible	Office of Zoonosis and Vector Control
	Reach	Beneficiaries	The beneficiary population encompasses all 16 regions of the country. There is no specific territorial prioritization, as the vectors and reservoirs prioritized by the measure could affect the entire territory; therefore, the scope is universal throughout the country.

			Further analysis of the distribution of vectors and reservoirs of public health importance in territories where their registration is not available.												
		Territorial	National												
Synergies	It incorporates or contributes to the cross-cutting guidelines of ECLP, PNACC, NDC	ECLP: Meta 4.3 - It is linked to the criterion: "Cost-efficiency: reduction of costs in the medium and long term" of the social pillar of the NDC The measure is linked to the response capacity of the healthcare network to vectors and reservoirs (measure No. 9 of the Plan) Not relevant to gender mainstreaming.													
	Synergy with other public policies	- Synergy with the SDGs: 3 on Good Health and Well-being, 13 on Climate Action.													
Planning	Timeline Implementation (Main activities that must be carried out to achieve the goal of the (Measurement)	Actions (Specify the name of the actions)	Years					Indicators Actions (quantitative or qualitative)					Institutions		Total cost by action
			Year 1 2025	Year 2 2026	Year 3 2027	Year 4 2028	Year 5 2029	Number re	Meta	Unit of measure	Medium of Verification	Categorization Media of Implementation (T, C, F u O)8	Responsible	Collaborators	
		Action 1 Design the system monitoring considering the control actions and communication.	x (2023 and 2024)					Memo robbed m near this of design	Design of the system	Number of Information of Design of the system	Memorandum sent to DEGREYD con design proposal	Development and transfer of technology.	Office of Zoonoses and Control of Vectors (DIPOL).		M\$8.745
		Cost	M\$8.745												
		Action 2 Development and Implementation of system	x (2024 and 2025)					Imple mint tion of the System a	System implementad or in MIDAS	Number of Document you officers	Ordinary to the SEREMIS of Health indicating that the system is operational	Creation and strengthening to de capabilities	Office of Zoonoses and Control of Vectors (DIPOL).	SEREMIS of Health Team MIDAS (ICT-MINSAL)	M\$98.364
		Cost	M\$98.364												
		Action 3 Maintenance and System operation		x	x	x	x	Inform and of Produ ction Annual of the System a MAC	Generate a consolidated annual report that summarizes the actions key and the results system operators MAC	Number of Information generated per year	Annual Report submitted to the authorities, supported by annexes, quarterly reports, data from monitoring and evaluation		Office of Zoonoses and Control of Vectors (DIPOL)	SEREMIS of Health Team MIDAS (ICT-MINSAL) CASE – ISP – Academia (Animal diagnostic laboratories) – SENAPRED, Department of Communications (MINSAL)	M\$90.792
		Cost		M\$22.698	M\$22.698	M\$22.698	M\$22.698								
		Total cost per year	M\$107.108	M\$22.698	M\$22.698	M\$22.698	M\$22.698								M\$197.900
Financing	Estimated cost of the measure (M\$)	M\$197.900													
	Possible sources of financing	To be defined by the Ministry of Health. Additional budget is required.													
Additional Information	Assumptions	- The first assumption is that the funding will be available to carry out the committed activities. - There are no threats of natural or anthropogenic origin of great magnitude and territorial coverage. - That the resources in qualified personnel and financial resources be maintained for the execution of the activities of the regional programs of the Zoonosis and Vector Control Units in the SEREMIS of Health.													

4.5.- Adaptation Measure No. 5: Strengthening the response capacity of the healthcare network for vectors and reservoirs

This measure was not included in the 2016-2022 Plan and is being incorporated into sector planning for the first time. Vector-borne diseases can be caused by parasites, bacteria, or viruses, and represent 17% of the estimated global burden of disease.

⁸ Indicate if the action can be categorized within any of the following means:

T: Technology development and transfer.

C: Capacity building and strengthening

F: Financial guidelines.

O: Other.

Infectious diseases cause more than 700,000 deaths annually worldwide. Due to climate change, an increase in the spread and prevalence of vector-borne diseases and infections is expected. This effect is not only directly caused by climate change, but also indirectly through population growth, migration, land use, poor urbanization, biodiversity loss, and other factors.

Regarding disease vectors in Chile, an increase in the population exposed to mosquitoes is expected; in fact, a greater presence of mosquitoes has already been observed on Rapa Nui and in the Arica and Parinacota and Tarapacá regions. Changes in the presence of rodents that transmit Hantavirus are expected in the central and southern regions, which could affect the incidence of this disease.

Measure No. 5: “Strengthening the response capacity of the healthcare network to vectors and reservoirs.”															
Element	Sub-element	Content													
Measurement	Aim	Reduce vulnerability to the spread of vectors and reservoirs.													
	Specific (OE)														
	Lineament														
	Strategic (LE)	E2. Analysis and Research on impacts and scenarios.													
	Description and result expected from the measure	Develop and maintain up-to-date response plans for the healthcare network in the event of outbreaks of vector-borne diseases and their reservoirs. These plans must be communicated to healthcare personnel through training sessions designed to strengthen the knowledge of healthcare teams in responding to the emergence and/or epidemics of these diseases. Response plans must be updated every three years as new information becomes available.													
	Indicator Measurement	1. Number of active vector-borne disease response plans / total number of plans 2. Number of updated guidelines, regulations and instructions for the care of vector-borne diseases / planned number 3. Percentage of trained officials / staff expected to be trained													
	Institutions	Responsible	Healthcare Network Management Division, Benefits												
Reach		Beneficiaries	healthcare teams in health facilities.												
		Territorial	At the national level												
Synergies	It incorporates or contributes to the cross-cutting guidelines of ECLP, PNACC, NDC	ECLP: Target 2.3 (By 2030, increase the national knowledge base in key health sector teams including primary care and higher levels of complexity. NDC A 2)													
	Synergy with other public policies	ENS: OE 7 Environment: IO 7.4 Reduce the negative impact on health due to climate change. OE: Transmissible Diseases: IO Decrease the incidence of zoonotic or vector-borne diseases.													
Planning	Timeline Implementation (Main activities that must be carried out to achieve the goal of the (Measurement)	Actions (Specify the name of the shares)	Years					Indicators Actions (quantitative or qualitative)					Institutions		Total cost by action
			Year 1 2025	Year 2 2026	Year 3 2027	Year 4 2028	Year 5 2029	Name	Of a	Unit of measure	Medium of Verification	Categorization Media Implementation (T, C, F u O) ⁹	Responsible for	Collaborators	
		Action 1 Develop and keep up-to-date the healthcare network's outbreak response plans of diseases transmitted by vectors and reservoirs.	Development of Diagnostic.	Develop hole of Plan of action				Report with Diagnosis and Plan	100 %		Documents in SISDOC	C: Capacity building and strengthening	DIGEST	DIVAP – Epi SP- DiPOL	M\$ 0
		Cost	M\$ 0	M\$ 0 M\$ 0		M\$ 0 M\$ 0									
		Action 2 Dissemination of instances of Education and training for health teams on the presence of vectors and reservoirs in Chile associated with diseases such as malaria, dengue, rabies, among others, and the diagnosis, treatment and response plans for these diseases			Develop capacity hole tacion is.			Percentage of trained healthcare teams,	30 %			C: Capacity building and strengthening	DIGEST	DIVAP – Epi SP- DiPOL	M\$ 0
		Cost	M\$ 0	M\$ 0 M\$ 0 M\$ 0 M\$ 0											

⁹ Indicate if the action can be categorized within any of the following means:

T: Technology development and transfer.

C: Capacity building and strengthening

F: Financial guidelines.

O: Other.

		Total Cost	MS 0	MS 0	MS 0	MS 0	MS 0								MS 0
-	Estimated cost of the measure	The development of the plans and training will be carried out using the human resources of the institutional staff and budget.													
—	Possible sources of funding	Regular budget of the Ministry of Health.													
—	Assumptions	For the development of the Action Plan, there will be budgetary availability for the measures that require financing.													

4.6.- Adaptation Measure No. 6: Strengthening of human health capital public

The ACE is an internationally conceived instrument to promote climate action that is included in article 6 of the UNFCCC, referring to education, training and public awareness, in its component 2 it points to the creation and strengthening of capacities for climate change management.

Climate change and its impacts on health pose significant challenges to the health sector's institutional framework in responding to and addressing the new needs associated with climate change. It is crucial to ensure the technical capacity of human resources to recognize and address the demands related to its manifestations, its effects on the health and safety of individuals, and potential changes in the population's morbidity and mortality profile. This includes emerging or re-emerging diseases such as dengue, rabies, hantavirus pulmonary syndrome, and yellow fever; non-communicable diseases such as cardiorespiratory and allergic diseases; effects on mental and psychological health; and other impacts mediated by adverse environmental conditions and events associated with climate change. Strengthening the sector's human capital requires a coordinated, cross-cutting approach and effort that incorporates diverse perspectives, including epidemiology, risk management, gender equality, intersectoral collaboration, and cultural sensitivity, among others.

During the period of the 2017-2022 adaptation plan, regional training on climate change was incorporated. This measure considered both the central and regional levels, achieving 54% progress during this period. Therefore, this measure aims to complete the work begun in 2017 and reach the primary healthcare level.

To achieve this objective, courses will be conducted for Regional Ministerial Secretariats and Health Services, and subsequently, digital platforms will be used to disseminate information widely in Primary Health Care. The content will include the following:

They will be associated with adaptation to climate change, impacts on human health, the need for coordinated joint response to epidemics, and efficient use of resources.

Measure No. 6: Strengthening public health human capital		
Element	Sub-element	Content
of Description	Aim Specific (OE)	Strengthening human capital in health regarding the relationships between climate change and impacts on human health to highlight the importance of preparing to face them and building resilience.
	Lineament Strategic (LE)	E1 Health Education, training, awareness and cooperation.

	Description and result expected from the measure	Health sector personnel (central level of the Undersecretariats of Public Health and Healthcare Networks, regional level of the Regional Ministerial Secretariats and Health Services, and municipal health services) trained on climate change and its expected impacts at the national, regional, and local levels, as well as trained education personnel. Identifying key officials to whom the training would be directed and how to replicate it in their respective sectors. GOAL: To strengthen human capital in health in relation to the interactions between climate change and impacts on human health, highlighting the importance of preparedness and building resilience.													
	Indicator Measurement	1. Training design. 2. Percentage of centrally trained staff/number of expected trainees. 3. Percentage of staff trained at the Seremi and Health Services level / number of people trained planned by Seremi and Health Services. 4. Percentage of trained APS level/ number of planned trainees.													
	Institutions	Responsible	Ministry of Health Degreyd Central level: - SSSP: Epidemiology. Office of Environmental Epidemiological Surveillance, Surveillance of Communicable Diseases, Zoonoses and Vectors, Environmental Health, Nutrition and Food, Communicable Diseases, Non-Communicable Diseases, Occupational Health. - SSRA: Healthcare Risk Management, Model, Processes and Hospital Transformation, Health Services Management, Centralized Bed Management, National Coordination of Drugs and Supplies, DIVAP. - GABMIN: Department of Human Rights and Gender - Public Health Institute (ISP) Regional level: - Regional Ministerial Secretariats of Health. - Health Services. - Municipalized Health. - Representatives of the Regional and Municipal Governments. Healthcare network: - Primary care staff. - Staff at higher levels of complexity. Collaborators: Regional Governments, Municipalities, MINEDUC.												
	Reach	Beneficiaries	It benefits health services and primary care staff. Reach: To provide knowledge to health personnel at the central, regional and healthcare network levels to address the impacts of climate change. More efficient and effective attention to the effects of climate change on prevention and people's health. Having health and education personnel trained in the impact of climate change on health.												
Synergies	It incorporates or contributes to the cross-cutting guidelines of ECLP, PNACC, NDC	PACCSS1: Measures 2 and 3 ECLP: Meta 2.3 ENS: LT1 - Active participation: the aim is to transmit knowledge about climate change and health to health personnel at the central, regional and healthcare network levels.													
	Synergy with other public policies	- Synergy with the SDGs: 3 on Good Health and Well-being, 13 on Climate Action. Gender responsiveness: Promoting gender-equal participation in educational activities.													
Planning	Timeline Implementation (Main activities that must be carried out to achieve the goal of the (Measurement)	Actions (Specify the name of the shares)	Years					Indicators Actions (quantitative or qualitative)					Institutions		Total cost per action
			Year 1 2025	Year 2 2026	Year 3 2027	Year 4 2028	Year 5 2029	Number re	Meta	United d of measure a	Medium of Check us out	Categorization Media of Implementacion (T, C, F u O)10	Answers	Collaborate time	
		Action 1 Train key personnel and teams at the level of the central Ministry of Health.	Level course Central					Cover itation Level Center i	100	%	Report Biannual	C: Creation of capabilities	Degreyd	Services Health M\$ 11,507	
		Cost	M\$ 11,507	M\$ 0	M\$ 0	M\$ 0	M\$ 0								
		Action 2 To train key personnel and teams at the regional level of SEREMIS and Services of Health	Design of the training program	Courses coach is	Tutoring for APS	Tutoring for APS	Tutoring for APS	Cover itation Seremi s	100	%	Report Biannual	C: Creation of capabilities	Degreyd	Services Health M\$ 20,958	
		Cost	M\$ 3,159	M\$17,783 M\$ 0		M\$ 0	M\$ 0								
		Action 3 Train key personnel and teams in Care Primary	Design of the training program	Courses online	Online courses	Online courses		Capable tion APS	80	%	Report Biannual	C: Creation of capabilities	Degreyd	APS	M\$ 18,760
		Cost	M\$ 3,159	M\$ 3,159	M\$ 3,900	M\$ 3,900	M\$ 3,900								

¹⁰ Indicate if the action can be categorized within any of the following means:

T: Technology development and transfer.

C: Capacity building and strengthening

F: Financial guidelines.

O: Other.

		Total cost per year	M\$ 17.825	M\$ 21.678	M\$ 3.900	M\$ 3.900	M\$ 3.900							M\$ 51.205
	Estimate of cost of the measure	M\$ 51.206												
Financing	Possible sources of funding	To be defined by the Ministry of Health. Additional budget is required.												
additional Information	Assumptions	Development of training tools, leveraging national and international expertise in health and climate change, considering, as appropriate, key training themes such as gender, intersectionality, and cultural and territorial relevance, among others. Implementation of training programs at all levels. The necessary budget and staff availability will be ensured. This would be a Train the Trainer training and they will be the tutors at the regional level for the online courses and certifications of Primary Health Care staff. Staying at the institution.												

4.7.- Adaptation Measure No. 7: Epidemiological surveillance system of effects on health associated with climate change

Climate change has been described as a growing threat to human health, as it could increase the likelihood of serious impacts on the health of the population, mainly in those effects related to extreme heat events, vector diseases and diseases transmitted by food and water; in addition, it has been described that these phenomena can affect the social determinants of health in a generalized and irreversible way (Cagua, A. 2017).

Determining the impact of these events on population health through epidemiological surveillance is essential for developing public health policies aimed at reducing the risk of adverse health effects associated with climate change. In this regard, the development of surveillance and monitoring systems would not only provide insight into the population's health status but also generate evidence for clinical research and strengthen communication channels between the central government and the regions.

The greatest challenge will be maintaining enhanced and adaptable epidemiological surveillance and monitoring changes in the epidemiological patterns of communicable and non-communicable diseases caused or exacerbated by climate change, assessing risks to anticipate them. This surveillance can be complemented by mathematical models that include climate variables, among others, to more accurately predict disease transmission patterns, as well as by geospatial viewers.

In the context of the Climate Change Adaptation Plan, the Department of Epidemiology has arranged the following actions:

Adaptation Measure N 7: "Epidemiological surveillance system for health effects associated with climate change"		
Element	Sub-element	Content
Measurement	Aim	Develop an integrated and unified system for registering health events associated with climate change and an epidemiological surveillance system for health effects associated with climate change.
	Specific (OE)	
	Lineament	E2 Impact and Scenario Analysis and Research
	Strategic (LE)	
	Description and expected result of the measure	<u>Description:</u> Epidemiological surveillance of effects on human health associated with climate change, extreme temperature events and monitoring of indicators: systematic analysis of existing information sources such as hospital discharge and death databases of the Department of Statistics and Health Information (DEIS) for diseases associated with extreme temperature events, vector diseases, foodborne diseases (whose investigation provides possible relationships with climatic variables) or others. <u>Expected result:</u> - To have a registration system for the epidemiological surveillance of communicable and non-communicable diseases associated with climate change. - Having a methodological tool for the development and updating of risk matrices. - Having an annual analysis of emergency room visits, hospital discharges and mortality from diseases associated with climate change (associated with extreme heat, extreme temperatures, heat waves, extreme weather events), vector-borne diseases, waterborne diseases*, foodborne diseases*, emergency room visits during critical episodes/

		Environmental pre-emergency in saturated areas; forest fires; algal bloom episodes. - Alerting to epidemiological changes in communicable and non-communicable diseases that are affected by climate change. - Adapt the surveillance system to these new risks if necessary. - To provide information for decision-making with an emphasis on prevention, reducing hospitalization and mortality *When they are associated with climate change. Implement and consolidate an integrated and unified system for registering health events associated with climate change, accompanied by an epidemiological surveillance system that allows monitoring, analysis and alerting about the health effects related to extreme temperatures, vector-borne diseases, water and food-borne diseases, as well as other extreme weather events, providing key information for public health decision-making.																																																																				
	Indicator Measurement	1. Annual reports on ENO associated with climate change (vector-borne diseases, waterborne diseases, foodborne diseases, respiratory diseases) 2. Number of epidemiological alerts / events that occurred. 3. Number of risk assessments performed / assessments planned 4. Annual Report on the Hospital Discharge Rate for Non-Communicable Diseases Attributable to Climate Change (associated with extreme heat, extreme temperatures, heat waves, extreme weather events). 5. Annual report with the mortality rate from non-communicable diseases attributable to climate change. 6. Number of emergency room visits for illnesses attributable to climate change, extreme temperatures, heat waves and extreme weather events, critical episodes of environmental emergency/ pre-emergency in saturated areas; forest fires; algal bloom episodes.																																																																				
	Institutions	<table><tr><td>Responsible</td><td>Department of Epidemiology. Office of Surveillance of Communicable Diseases and Office of Environmental Epidemiological Surveillance.</td></tr></table>	Responsible	Department of Epidemiology. Office of Surveillance of Communicable Diseases and Office of Environmental Epidemiological Surveillance.																																																																		
Responsible	Department of Epidemiology. Office of Surveillance of Communicable Diseases and Office of Environmental Epidemiological Surveillance.																																																																					
	Reach	<table><tr><td>Beneficiaries</td><td> The health authority, through the strengthening of its capacities. The result of the measure will benefit the population of the entire country. Development and strengthening of surveillance and monitoring systems, which will allow us to know not only the health status of the population, but also to generate evidence for clinical research and strengthen communication channels between the central level and the regions. </td></tr><tr><td>Territorial</td><td>National, Regional, Municipal SOS</td></tr></table>	Beneficiaries	The health authority, through the strengthening of its capacities. The result of the measure will benefit the population of the entire country. Development and strengthening of surveillance and monitoring systems, which will allow us to know not only the health status of the population, but also to generate evidence for clinical research and strengthen communication channels between the central level and the regions.	Territorial	National, Regional, Municipal SOS																																																																
Beneficiaries	The health authority, through the strengthening of its capacities. The result of the measure will benefit the population of the entire country. Development and strengthening of surveillance and monitoring systems, which will allow us to know not only the health status of the population, but also to generate evidence for clinical research and strengthen communication channels between the central level and the regions.																																																																					
Territorial	National, Regional, Municipal SOS																																																																					
Synergies	Incorporates or contributes to the guidelines cross-sections of ECLP, PNACC, NDC	This measure contributes to the cross-cutting guidelines of ECPL, PNACC, and NDC regarding identifying vulnerabilities, providing information for health decision-making, and protecting people's health. It contributes to strengthening and improving the governance of health information and human systems, identifying specific vulnerable groups, and building and strengthening capacities.																																																																				
	Synergy with other public policies	Alignment with other public policies: National Health Strategy, Strategic Objective No. 1. Communicable Diseases (Zoonotic and Vector-borne Diseases and other communicable diseases) and Strategic Objective No. 7 "Environment" (Sanitary-environmental conditions and Climate Change Management for Health). <u>Synergy with SDGs: 3 on Good Health and Well-being and 13 on Climate Action.</u> Level of gender mainstreaming: The measure considers differences in health impacts according to gender. This measure will provide information to develop public health policies aimed at reducing the risk of people experiencing adverse health effects associated with climate change.																																																																				
Planning	Timeline Implementation (Main activities that must be carried out to achieve the goal of the (Measurement)	<table><tr><th rowspan="2">Actions (Specify the name of the actions)</th><th colspan="5">Years</th><th colspan="5">Indicators Actions (quantitative or qualitative)</th><th colspan="2">Institutions</th><th rowspan="2">Total cost per action</th></tr><tr><th>Year 1</th><th>Year 2</th><th>Year 3</th><th>Year 4</th><th>Year 5</th><th>Name</th><th></th><th>Meta</th><th>Unit of measure</th><th>Medium of Verification</th><th>Categorization Media Implementation (T, C, F u O)¹¹</th><th>Responsible for Collaborators</th></tr><tr><td>Action 1: Epidemiological surveillance of diseases transmissible and non-transmissible diseases associated with climate change, information analysis and preparation of information</td><td>x</td><td></td><td>x</td><td></td><td></td><td>x</td><td>Report epidemiological surveillance logic of disease Health effects address transferable them and not transmissible the associates in return climatic.</td><td></td><td>Number of Document of the officers</td><td>Epidemiological surveillance reports as of illness + transmissible and non-communicable diseases associated with change climatic**</td><td>Other</td><td> Department to de Epidemiology a. Office of Surveillance of Disease Transmissible Diseases and Epidemiological Surveillance Office Environmental ca. MMA (¹² Academia, Address Meteorological Chile (DMC) It is expected that the Institutions collaborators can provide information about their competence when it is required, to supplement the information </td><td>M\$56.865</td></tr><tr><td>Cost</td><td>M\$ 18.955</td><td>M\$</td><td>M\$ 18.955</td><td>M\$</td><td></td><td>M\$18.955</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Action 2: Development of risk maps for communicable and non-communicable diseases transmissible associated with change climate for the country.</td><td></td><td></td><td>x</td><td></td><td></td><td>x</td><td>Risk maps for each disease ad transmissible law and no transmissible the associate in return climate, considers</td><td></td><td>Number of Document you officers</td><td>Risk maps for diseases transmissible and non-transmissible associated with change climatic, considered in this document</td><td>Other</td><td> Department the of Epidemiology a. Disease Surveillance Office + Transmissible and the Office of Epidemiological Surveillance Environmental ca. </td><td>M\$6.000</td></tr></table>	Actions (Specify the name of the actions)	Years					Indicators Actions (quantitative or qualitative)					Institutions		Total cost per action	Year 1	Year 2	Year 3	Year 4	Year 5	Name		Meta	Unit of measure	Medium of Verification	Categorization Media Implementation (T, C, F u O) ¹¹	Responsible for Collaborators	Action 1: Epidemiological surveillance of diseases transmissible and non-transmissible diseases associated with climate change, information analysis and preparation of information	x		x			x	Report epidemiological surveillance logic of disease Health effects address transferable them and not transmissible the associates in return climatic.		Number of Document of the officers	Epidemiological surveillance reports as of illness + transmissible and non-communicable diseases associated with change climatic**	Other	Department to de Epidemiology a. Office of Surveillance of Disease Transmissible Diseases and Epidemiological Surveillance Office Environmental ca. MMA (¹² Academia, Address Meteorological Chile (DMC) It is expected that the Institutions collaborators can provide information about their competence when it is required, to supplement the information	M\$56.865	Cost	M\$ 18.955	M\$	M\$ 18.955	M\$		M\$18.955								Action 2: Development of risk maps for communicable and non-communicable diseases transmissible associated with change climate for the country.			x			x	Risk maps for each disease ad transmissible law and no transmissible the associate in return climate, considers		Number of Document you officers	Risk maps for diseases transmissible and non-transmissible associated with change climatic, considered in this document	Other	Department the of Epidemiology a. Disease Surveillance Office + Transmissible and the Office of Epidemiological Surveillance Environmental ca.	M\$6.000
		Actions (Specify the name of the actions)		Years					Indicators Actions (quantitative or qualitative)					Institutions			Total cost per action																																																					
			Year 1	Year 2	Year 3	Year 4	Year 5	Name		Meta	Unit of measure	Medium of Verification	Categorization Media Implementation (T, C, F u O) ¹¹	Responsible for Collaborators																																																								
		Action 1: Epidemiological surveillance of diseases transmissible and non-transmissible diseases associated with climate change, information analysis and preparation of information	x		x			x	Report epidemiological surveillance logic of disease Health effects address transferable them and not transmissible the associates in return climatic.		Number of Document of the officers	Epidemiological surveillance reports as of illness + transmissible and non-communicable diseases associated with change climatic**	Other	Department to de Epidemiology a. Office of Surveillance of Disease Transmissible Diseases and Epidemiological Surveillance Office Environmental ca. MMA (¹² Academia, Address Meteorological Chile (DMC) It is expected that the Institutions collaborators can provide information about their competence when it is required, to supplement the information	M\$56.865																																																							
		Cost	M\$ 18.955	M\$	M\$ 18.955	M\$		M\$18.955																																																														
Action 2: Development of risk maps for communicable and non-communicable diseases transmissible associated with change climate for the country.			x			x	Risk maps for each disease ad transmissible law and no transmissible the associate in return climate, considers		Number of Document you officers	Risk maps for diseases transmissible and non-transmissible associated with change climatic, considered in this document	Other	Department the of Epidemiology a. Disease Surveillance Office + Transmissible and the Office of Epidemiological Surveillance Environmental ca.	M\$6.000																																																									

¹¹ Indicate if the action can be categorized within any of the following means:

T: Technology development and transfer.

C: Capacity building and strengthening

F: Financial guidelines.

O: Other.

¹²The MMA is expected to be able to collaborate with the transfer of information.

								days in it is a document							
		Cost	MS2.000 MS 0		MS2.0 00	MS 0	MS2.0 00								
		Total cost per year	MS20.955 MS 0		MS20. 955	MS 0	MS20. 955								MS62.865
Financing	Estimated cost of the measure	Surveillance of diseases associated with climate change, information analysis and report preparation, for 6 months \$56,865,384 Development of risk maps for climate change-related diseases in the country, using information generated through epidemiological surveillance \$6,000,000 Total \$62,865,384 (sixty-two million eight hundred sixty-five thousand three hundred eighty-four), Chilean pesos.													
	Possible sources of financing	To be defined by the Ministry of Health. Additional budget is required. Possible support from PAHO.													
Information additional	Assumptions	- Sufficient funding is available to achieve the objective of the measure and to carry out the proposed activities - Up-to-date data and information are available for epidemiological surveillance.													

4.8.- Adaptation Measure No. 8: Increase in the number of services that can be provided online in the health sector to avoid exposure to climate risks

This measure was not included in the 2016–2022 Health Sector Plan and was incorporated for the first time in the 2025–2029 plan. Due to climate change, extreme weather events will become increasingly frequent, and a significant increase in the incidence of morbidity and premature mortality associated with climate-sensitive diseases is expected. On both the hottest and coldest days, above the average summer temperature, morbidity and mortality are known to increase due to compromised body temperature regulation or the induction of direct and indirect health complications.

This measure aims to strengthen remote care strategies to ensure continuity of care during extreme weather events. Its objective is to increase the resilience of the healthcare system by guaranteeing continuity of care and seeking to reduce disruptions.
the risk of exposure, with a focus on populations that are more vulnerable to these events, such as children, the elderly, and those with chronic illnesses.

Measure No. 8: “Increasing the number of online health services available to avoid exposure to climate risks.”				
Element	Sub-element	Content		
Measurement	Specific Objective (ARE YOU)	Reduce the population's exposure to extreme weather events.		
	Lineament Strategic (LE)	E6. Climate risk reduction		
	Description and expected result of the measure	Strengthening online services to reduce the exposure of vulnerable populations to extreme weather events, increasing the number of digitized health services, which can include issuing prescriptions for chronic patients, requesting medical appointments, well-child check-ups for children and adolescents, requesting tests, telemedicine, and home delivery of medications.		
	Indicator Measurement	1. Health services adapted for online care (issuing prescriptions, requesting medical appointments, well-child check-ups for children and adolescents, requesting tests, etc.), / the total benefits. 2. Number of consultations carried out via telemedicine (e.g., Digital Hospital) during extreme weather events / the average of consultations on days without extreme weather events.		
	Institutions	Responsible	Healthcare Network Management Division.	
	Reach	Beneficiaries	Benefits Users of the Public Health System	
		Territorial	At the national level	
the Synergies	Incorporates or contributes to the guidelines	Synergy with the SDGs: 3 on Good Health and Well-being, 13 on Climate Action		

	cross-sections of ECLP, PNACC, NDC														
	Synergy with other public policies	ENS: OE 7 Environment - IO 7.1 Reduce the population exposed to unfavorable health and environmental conditions. IO 7.4 Reduce the negative impact on health due to climate change.													
Planning	Timeline Implementation (Main activities that must be carried out to achieve the goal of the (Measurement))	Actions (Specify the name of the shares)	Years					Indicators Actions (quantitative or qualitative)					Institutions		Total cost per action
			Year 1 2025	Year 2 2026	Year 3 2027	Year 4 2028	Year 5 2029	Name	Meta	Unit of measurement	Medium of Check us out	Categorization Media of Implementation (T, C, F u O)13	Responsible	Contributor	
		Action 1 Develop a diagnosis of possible services that can be performed online.	Development of Diagnostic					Document them Reports of Diagnosis co.	100%	Number of reports.	Zero Paper	C: Creation and strengthening to of capabilities	DIGEST	Division Attention Primary – Digital Health - TIC	MS 0
		Cost	MS 0	MS 0 MS 0		MS 0 MS 0									
		Action 2 Development of a migration plan from in-person to online		Develop hole in the plan	Imple mint sion of the Plan			Chronograph report my simple ntation	80%	%	Pan cumplmie	C: Creation and strengthening to de capabilities	DIGEST	Division Attention Primary – Digital Health - TIC	MS 0
		Cost	MS 0	MS 0 MS 0	MS 0 MS 0										
		Action 3 Evaluate the effect of the plan to migrate benefits to an online modality.					Evaluation	Performance report nes that have migration a modality d online by the year 2028.	100%	Report No.	Zero Paper	C: Creation and strengthening to de capabilities	DIGEST	Division Attention Primary – Digital Health - TIC	MS 0
		Cost	MS 0	MS 0 MS 0		MS 0 MS 0									
		Total Cost Per Year	MS 0	MS 0 MS 0		MS 0 MS 0									MS 0
Financing	Estimated cost of the measure	The proposed actions will be developed using the institutional human resources and budget, coordinating the different divisions and departments involved.													
	Possible sources of financing	Regular budget of the Ministry of Health.													
Informational	Assumptions	There is collaboration from ICT and Digital Health for the development of actions and tasks.													

4.9.- Adaptation Measure No. 9: Development of a response plan for the healthcare network to heat waves

Due to climate change, a significant increase in the incidence of morbidity and premature mortality associated with climate-sensitive diseases is expected. Heat waves, in particular, are associated with an increased incidence of health problems such as strokes, heart failure, and other heat-related conditions, such as dehydration and heatstroke. These conditions pose a greater risk to vulnerable populations, such as children, older adults, patients with cardiovascular disease, and people who work outdoors. The WHO warns that high temperatures can lead to increased levels of anxiety, stress, aggression, dehydration, and heatstroke, affecting both physical and psychological health.

Given this scenario, the need arises for clear protocols on the response of the healthcare network to this specific phenomenon, which justifies the development of this measure.

¹³ Indicate if the action can be categorized within any of the following means:

T: Technology development and transfer.
C: Capacity building and strengthening
F: Financial guidelines.
O: Other.

This measure was not part of the 2016-2022 Sectoral Health Plan and, in light of the above, its incorporation is necessary.

At the same time, the infrastructure of health facilities must be adapted to face high temperature phenomena, implementing measures that ensure adequate environmental conditions for care, such as temperature control, ventilation and humidity.

Measure No. 9: “Development of a response plan for the healthcare network to heat waves.”															
Element	Sub-element	Content													
Measurement	Specific Objective (ARE YOU)	Reducing the vulnerability of the healthcare network to heat waves.													
	Lineament Strategic (LE)	E4. Climate risk reduction.													
	Description and expected result of the measure	Strengthen the response capacity of the Healthcare Network through the implementation of protocols and the adaptation of infrastructure to reduce the vulnerability of health centers to heat waves.													
	Indicator Measurement	1. Design of the Ministry of Health's response plan. 2. Percentage of Health Services that have heat wave response protocols / total number of Health Services 3. Number of training sessions to disseminate guidelines related to care during heat wave events / number of training sessions planned. 4. Number of officials trained in the application of healthcare response strategies to heat waves / total primary healthcare staff expected to train.													
	Institutions	Responsible	Healthcare Network Management Division, Primary Care Division, Investment Division. Benefits healthcare teams in health												
	Reach	Beneficiaries	facilities and users of the public health network.												
		Territorial	At the national level												
Synergies	It incorporates or contributes to the cross-cutting guidelines of ECLP, PNACC, NDC	ECLP: Target 2.3 (By 2030, increase the national knowledge base in key health sector teams), ENS: SO 7 Environment. OI 7.1. Reduce the population exposed to unfavorable health and environmental conditions. OI 7.4. Reduce the negative health impact of climate change. SO 8 Health Systems. OI 8.10 Mitigate the effects of emergencies and disasters on health and well-being.													
	Synergy with other public policies	ECLP: Meta 1.2, Meta 2.3, ENS. OE 7 y OE 8.													
Planning	Timeline Implementation (Main activities that must be carried out to achieve the goal of the (Measurement)	Actions (Specify the name of the shares)	Years					Indicators Actions (quantitative or qualitative)					Institutions		Total cost per action
			Year 1 2025	Year 2 2026	Year 3 2027	Year 4 2028	Year 5 2029	Name	Meta	Unit of measure	Medium of Check us out (Indicate)	Categorization Media of Implementation (T, C, F or O) ¹⁴	Response able	Collaborate door	
		Action 1 Develop and disseminate a “Healthcare response plan for pathologies associated with waves of heat”, compatible with the response plan adopted by the Ministry of Health within the framework of the work of SENAPRED.	Design of the Plan and approval of action (Indicate)	Act linking on and diffusion (Indicate)	Act lizació ny difusi (Indicate)			Documents incorporating healthcare response to pathologies associated with heat waves.	100	%	Dissemination of the Plan of Response from Ministry Health (Indicate)	C: Creation and strengthening into de capabilities	DIGEST	Division Attention Primary	M\$ 0
		Cost Action 2 Development of a protocol with measures to respond to extreme high temperature events and heat waves in the infrastructure of health, to ensure the conditions environmental such as, temperature, ventilation, humidity, for healthcare activity	Design of the Protoc solo and approve action (Indicate)	Act linking on and diffusion (Indicate)	Act lizació ny difusi (Indicate)			Specific protocol to ensure conditions environmental appropriate in the infrastructure day.	100	%	Diffusion of the Protocol of answer (Indicate)	C: Creation and strengthening into de capabilities	Division of Inversion is	DIGEST	M\$ 0
		Cost	M\$ 0	M\$ 0	M\$ 0	M\$ 0	M\$ 0								
		Total cost per year	M\$ 0	M\$ 0	M\$ 0	M\$ 0	M\$ 0								
Cost estimate of the measure	Reorient the role of professionals at the central level														

¹⁴ Indicate if the action can be categorized within any of the following means:

T: Technology development and transfer.

C: Capacity building and strengthening

F: Financial guidelines.

O: Other.

	Possible sources of financing	Regular budget of the Ministry of Health.
Informational	Assumptions	Human resources are available.

4.10.- Adaptation Measure No. 10: Development of a methodology that allows the preparation of regional emergency plans considering climate threats along with other local threats

Climate threats that can affect public health are diverse, and their relevance varies geographically. Local planning to address these threats requires a methodological tool that allows for the identification of current and future climate threats relevant to the local area, and then for their importance to be weighed against other non-climate threats in quantifiable comparative terms, considering criteria such as the probability of occurrence at the regional and local levels, the severity of the damage they could cause to the exposed population, and the degree of exposure of the population should they occur.

Traditionally, regional levels have developed their emergency plans using risk matrices where the recognition of local climate threats has not had a uniform methodological approach and the excessive aggregation of these threats has not always facilitated the comparative weighting of risk in the process of formulating or updating emergency plans; shortcomings that are expected to be overcome with this Adaptation Measure.

This measure was included in the 2017-2022 adaptation plan, and given the high turnover of staff and authorities, as well as the new climate events occurring in different regions, constant analysis and understanding of the various events that may affect health centers and the surrounding population are required. In this regard, Law No. 21,364 establishes that each municipality must have an updated emergency plan, which is developed taking into account the risk factors identified in the risk matrix analysis.

These matrices are based on the AHP (multi-criteria decision analysis) model, which allows for a more accurate assessment and comparison between the different threats that regions may be affected by.

Measure No. 10: "Implement a methodology that allows the development of regional emergency plans considering climate threats along with other local threats."		
Element	Sub-element	Content
Measurement	Aim Specific (OE)	Facilitate the management of health risks associated with prioritized climate threats with local relevance
	Lineament Strategic (LE)	E4. Climate risk reduction.
	Description and result expected from the measure	Implement and keep regional risk matrices up to date. This will allow for the local integration of climate threats into the formulation of regional/local emergency plans, ensuring adequate preparedness and timely response to emergencies and disasters. A specially designed methodological tool will be used to keep risk matrices updated with data from monitoring and surveillance. These risk matrices will be developed as the necessary background information is identified in the studies proposed within the PACCSS activities.

		GOAL: - Strengthened regional emergency plans using the methodological tool that allows the consideration of climate threats with local relevance.														
	Indicator Measurement	1. Training of regional staff to carry out regional risk matrix 2. Percentage of risk plans implemented (with incorporated risk matrix) or updated at the national level / national total														
	Institutions	Responsible	Ministry of Health Degreyd Contributors: MMA15, Meteorological Directorate of Chile													
	Reach	Beneficiaries	Benefits the health authority and the Health Services Environmental health, nutrition and food, epidemiology, zoonoses and vectors Scope: Strengthening the capacities of the health authority and the Health Services.													
		Territorial	At the national level													
Synergies	Incorporates or contributes to the guidelines cross-sections of ECLP, PNACC, NDC	ECLP: Meta 1.2, Meta 2.3, ENS: LT1														
	Synergy with other public policies	- Synergy with the SDGs: 3 on Good Health and Well-being, 13 on Climate Action Gender sensitive: making visible/explicit the differences in impacts by sex and gender in the selection of monitored factors.														
Planning	Timeline Implementation (Main activities that must be carried out to achieve the goal of the (Measurement)	Actions (Specify the name of the shares)	Years					Indicators Actions (quantitative or qualitative)					Institutions		Total cost per action	
			Year 1 2025	Year 2 2026	Year 3 2027	Year 4 2028	Year 5 2029	Name	Meta	Unit of measure	Medium of Verification	Categorization Media Implementation (T, C, F u O) ¹⁶	Answers Answers	Collaborate time		
		Action 1 <i>Course to complete and update the risk matrices</i>	x					Matrices risk	100 %		Certificate and	C: strengthen the capabilities	DEGREYD	Authority Sanitary and Health Services	M\$ 18.293	
		Cost	M\$ 18.293	M\$ 0	M\$ 0	M\$ 0	M\$ 0									
		Action 2 <i>Update matrices</i>		x	x	x	x	Actualizac ion of matrices risk	100 %		Plans for updated risk	C: strengthen the capabilities	DEGREYD	Authority Sanitary and Health Services	M\$ 0	
		Cost	M\$ 0	M\$ 0	M\$ 0	M\$ 0	M\$ 0									
		Total cost per year	M\$ 18.293	M\$ 0	M\$ 0	M\$ 0	M\$ 0								M\$ 18.293	
Financing	Estimate of cost of the measure	M\$ 18.293														
	Possible sources of financing	To be defined by the Ministry of Health. Additional budget is required.														
additional information	Assumptions	Risk matrices should be designed as the necessary background information is identified in the studies proposed in the PACCSS activities. Retention of the staff who will be trained with the AHP method for the preparation of the matrices and their incorporation into the Sectoral plan.														

4.11.- Adaptation Measure No. 11: Diagnosis of lifelines and threats climate in health facilities

The increasing frequency and intensity of extreme weather events makes it crucial to assess and improve the resilience of healthcare facilities. In this context, the ECLP (Emergency Climate Change Strategy) addresses adaptation to extreme weather events, such as heat waves. This climatic phenomenon poses a significant risk to public health, and the ECLP includes specific measures to mitigate these effects. For example, it promotes the implementation of early warning systems,

¹⁵ MMA collaboration for the transfer of updated information regarding any regional risks they may have.

¹⁶ Indicate if the action can be categorized within any of the following means:

T: Technology development and transfer.

C: Capacity building and strengthening

F: Financial guidelines.

O: Other.

Measure No. 11: A. “Diagnosis of vital lines in health establishments to face extreme weather events”															
Element	Sub-element	Content													
Measurement	Aim	Diagnosis of the vital lines of health facilities and their capacity to respond to extreme weather events													
	Specific (OE)														
	Lineament	E5. Vulnerability assessment and preparedness for climate events.													
	Strategic (LE)														
	Description and expected result of the measure	A survey of essential services will be conducted using official channels, such as the Emergency Module of the MIDAS platform (Modernization of Digital Information of the Health Authority). The aim is to gather information on response capacity (status, autonomy, capabilities) related to essential services such as water, electricity, medical gases, telecommunications, and fuel, taking into account the climate-related threats present at the locations of the following health facilities: Hospitals: Low complexity - Medium complexity - High complexity; Primary Health Care Facilities: Family Health Centers (CESFAM) - Community Family Health Centers (CECOSF) - Primary Emergency Care Service (SAPU) - High-Resolution Primary Emergency Care Service (SAR) - Rural Primary Emergency Care Service (SUR) - Rural Health Posts (PSR) - Rural General Clinic (CGR); Prehospital Care: Emergency Medical Service (SAMU) Bases - SAMU Regulatory Center. GOAL: To have up-to-date information on the response capacity of the vital lines of the Hospital Network, Prehospital and Primary Health Establishments.													
	Indicator	1. Health facilities that have completed the collection of information on their lifelines and climate threats using the Emergency Module of the MIDAS platform / total number of facilities nationwide. (at the different levels requested)													
	Measurement														
Institutions	Responsible	Ministry of Health – Investment Division													
	Beneficiaries	Users of the public health system. Facilities, Municipal Health Management Entities, Health Services and Ministry of Health (knowledge of network capacities)													
	Territorial	National													
Synergies	Incorporates or contributes to the guidelines cross-sections of ECLP, PNACC, NDC	Identify how the measure contributes to one or more cross-cutting guidelines regarding:													
		<u>General:</u> a) Generation of science-based foundations. b) Water security. c) Losses, damages and costs of inaction. d) Co-benefits in mitigation or adaptation. e) Nature-based solutions. <u>Strengthening Governance:</u> f) Institutional framework and governance. g) M&E system. OE. 7 Environment, OI. 7.4 Reduce the negative health impact of climate change. OE. 8 Health System, OI 8.10 Mitigate the effects of emergencies and disasters on health.					<u>Human Systems:</u> h) Gender equality and equity. i) Indigenous Peoples and ancestral knowledge. j) Human mobility. k) Other specific vulnerable groups. l) Disaster Risk Management (DRM). <u>Means of Implementation:</u> m) Capacity building and strengthening. n) Development and transfer of technology. o) Financial guidelines.								
	Synergy with other public policies	Synergy with the SDGs: 3 on Good Health and Well-being, 13 on Climate Action. Alignment with other public policies													
Implementation Planning	Timeline	Actions (Specify the name of the shares)	Years					Indicators Actions (quantitative or qualitative)					Institutions		Cost total per share
			Year 1 2025	Year 2 2026	Year 3 2027	Year 4 2028	Year 5 2029	Name	Meta	Unit of measure	Medium of Verification	Categorization Media	Responsible	Collaborate time	

(Main activities that must be carried out to achieve the goal of the (Measurement)											Implementation (T, C, F u O)17			
	Action 1 Formal request for information to the establishments, with technical validation of Health Services of the country.	x					Official letter issued	100	%	Official letter signed and distributed.	O: GRD	Division of Investments	DIGEST DIVAP	\$ 0
	Cost	MS 0 MS 0		MS 0 MS 0 MS 0										
	Action 2 Identify territorially the main climate threats to consider capable of affecting lifelines.	x					Fact sheet updated capabilities in platform MIDAS	100	%	Check on the MIDAS Platform, emergency module.	O: GRD	Division of Investments	DEGREYD	\$0
	Cost	MS 0 MS 0	MS 0 MS 0 MS 0											
	Action 3 Gather information on response capacity (status, autonomy, capabilities) with respect to the lifelines of the establishments High hospitalizations, Medium Complexity and Low Complexity		x				Get up information on the capacity of response (state, autonomy, capabilities) with respect to the lifelines of the establishments High, Medium and Low Complexity hospitals.	100	%	Report 1. with results and Ordinary with Prioritized list of hospital establishments of high, medium and low complexity that require improving their response capacity, according to the information received.	O: GRD	Division of Investments	DIGEST	MS24.000
	Cost	MS 0	MS24.000	MS 0 MS 0 MS 0										
	Action 4 Gather information on the response capacity (status, autonomy, capabilities) with respect to the lifelines of the Establishments of APS. With support from DIVAP			x			Get up Information on response capacity (status, autonomy, capabilities) with respect to lifelines of the Establishments from APS. With support from DIVAP.	100	%	Report 2. with results and Ordinary with prioritized list of CESFAM establishments that require improving their capacity to answer.	O: GRD	Division of Investments	DIVAP	MS24.000
	Cost	MS 0 MS 0		MS24.000	MS 0 MS 0									
	Action 5 Gather information on the response capacity (status, autonomy, capabilities) with respect to the lifelines of the Establishments of APS. With support from DIVAP				x		Get up Information on response capacity (status, autonomy, capabilities) with respect to lifelines of the Establishments from APS. With DIVAP support	100	%	Report 3. with results and Ordinary with prioritized list of establishments CECOSF SAPU – SAR – ON – Health posts Rural (PSR) - Office General Rural, who require improving their response capacity	O: GRD	Division of Investments	DIVAP	MS24.000
	Cost	MS 0 MS 0		MS 0	\$ MS24.000	MS 0								
	Action 6 Gather information on the response capacity (status, autonomy, capabilities) of pre-hospital facilities regarding essential services. With support from DIGERA				x		Gather information on the response capacity (state, autonomy, capabilities) regarding the lifelines of pre-hospital establishments.	100	%	Report 4. with results and Ordinary with prioritized list of establishments pre-hospital care providers who need to improve their response capacity.	O: GRD	Division of Investments	DIGEST	MS24.000
	Cost	MS 0 MS 0 MS 0			\$ MS24.000	MS 0								
	Total cost per year	MS 0	MS24.000	MS24.000	MS48.000	MS 0								MS96.000

¹⁷ Indicate if the action can be categorized within any of the following means:

T: Technology development and transfer.

C: Capacity building and strengthening

F: Financial guidelines.

O: Other.

Financing	Cost estimate of the measure	M\$96,000 Associated costs - MINSAL human resources for systematization of the survey, diagnosis and preparation of reports with results and prioritization.
	Possible sources of funding	To be defined by the Ministry of Health. Additional budget is required.
Information	Assumptions	Lack of information regarding the survey conducted by the Health Services. Not having the funding for the required human resources. Emergencies and disasters that could delay or suspend the measure.

Measure No. 11: B. "Implementation of improvements to lifelines in health facilities to cope with extreme weather events"

Element	Sub-element	Content													
Measurement	Aim Specific (OE)	Diagnosis of the vital lines of health facilities and their capacity to respond to extreme weather events,													
	Lineament Strategic (LE)	E5. Vulnerability assessment and preparedness for climate events.													
	Description and expected result of the measure	Description of the Implementation Program: Based on the results reports and the prioritized list of establishments of measure No. 11.A, an implementation program will be carried out to improve health establishments that need to strengthen their response capacity in their vital lines, specifically drinking water and electricity. The Health Services must submit the projects developed according to the type of initiative, to be incorporated into the implementation program. The projects will cover the hospital network and the CESFAMs (Family Health Centers). This program will determine the funding sources and timelines associated with implementing the improvements. The evaluation of the continuation of these programs for the remaining Primary and Pre-Hospital Care facilities will be postponed until the first quarter of 2027. Ensure that the lifelines of prioritized health facilities, including hospitals and CESFAMs, are adequately prepared and improved to cope with extreme weather events, increasing their response capacity and resilience to these events.													
	Indicator Measurement	1. Number of health facilities (hospitals and CESFAMs) validated in programs that have implemented improvements in their vital lines / scheduled projects.													
	Institutions	Responsible	Health Services - Ministry of Health - Investment Division												
	Reach	Beneficiaries	Users of the public health system. Establishments within the hospital network and in the CESFAMs (Family Health Centers).												
		Territorial	National												
Synergies	Incorporates or contributes to the guidelines cross-sections of ECLP, PNACC, NDC	Identify how the measure contributes to one or more cross-cutting guidelines regarding: <u>General:</u> a) Generation of science-based foundations. b) Water security. c) Losses, damages and costs of inaction. d) Co-benefits in mitigation or adaptation. e) Nature-based solutions. <u>Strengthening Governance:</u> f) Institutional framework and governance. g) M&E system. OE. 7 Environment, OI. 7.4 Reduce the negative health impact of climate change. OE. 8 Health System, OI 8.10 Mitigate the effects of emergencies and disasters on health.							<u>Human Systems:</u> h) Gender equality and equity. i) Indigenous Peoples and ancestral knowledge. j) Human mobility. k) Other specific vulnerable groups. I) Disaster Risk Management (DRM). <u>Means of Implementation:</u> m) Capacity building and strengthening. n) Development and transfer of technology. o) Financial guidelines.						
	Synergy with other public policies	Synergy with the SDGs: 3 on Good Health and Well-being, 13 on Climate Action. Alignment with other public policies													
Planning	Timeline Implementation	Actions (Specify the name of the actions)	Years					Indicators Actions (quantitative or qualitative)					Institutions		Total cost by action
	(Main activities that must be carried out to achieve the goal of the (Measurement))		Year 1 2025	Year 2 2026	Year 3 2027	Year 4 2028	Year 5 2029	Goal Name		Unit of measure	Medium of Verification	Categorization Media Implementation (T, C, F u O) ¹⁸	Responsible	Contributor *	
		Action 1 Implementation program for improvements to lifelines of the establishments hospitals of High, Medium			X	X		Program tion of Implemen ation, The minute of program tion	100	16	Programming documents ny minuta	O: GRD	Division of Investments	DIGEST, Services of Health	

¹⁸ Indicate if the action can be categorized within any of the following means:

T: Technology development and transfer.

C: Capacity building and strengthening

F: Financial guidelines.

O: Other.

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4.12.- Adaptation Measure No. 12: Development of an internal plan of the Ministry of Health for the response to extreme high temperatures and heat waves

The extreme weather events with the highest mortality rates globally are related to periods of extreme high temperatures and the occurrence of heat waves. While our country's climate has some characteristics that generally mitigate the harmful effects of these events on public health, such as the significant difference between maximum and minimum temperatures on hot days and moderate relative humidity, two circumstances highlight the importance of having guidelines and protocols in place to prepare for and respond to these events: the vulnerability of the population to high temperatures is significantly greater in certain population groups, and the occurrence of extreme high temperature episodes and heat waves is becoming increasingly frequent, intense, and long-lasting across virtually the entire country.

The above highlights the need to develop an internal plan within the Ministry of Health for responding to extreme high temperatures and heat waves, which includes the competition

of all the technical areas involved, making coordinated use of the normative and guiding role that corresponds to the ministerial level.

The inclusion of new measures in an adaptation plan highlights that the previous plan focused on measures related to general health, but not specifically for the health sector. The inclusion of these new measures in the Ministry of Health's internal plan to respond to heat waves reflects an effort to adapt and improve public health strategies, ensuring they are more targeted and effective for the health sector.

One of the key areas of the ECLP is adaptation to extreme weather events, such as heat waves. Heat waves pose a significant risk to public health, and the ECLP includes specific measures to mitigate these effects. For example, it promotes the implementation of early warning systems, the improvement of urban infrastructure to reduce the impact of extreme heat, and the promotion of resilient agricultural practices.

Measure No. 12: “Development of an internal plan by the Ministry of Health for responding to extreme high temperatures and heat waves”																
Element		Sub-element		Content												
Description	Aim Specific (OE)		Reduce the vulnerability of the population to extreme high temperatures and heat waves.													
	Lineament Strategic (LE)		E4. Climate risk reduction.													
	Description and expected result of the measure		Develop and implement an internal plan of the Ministry of Health that prepares and coordinates all the Divisions and technical areas of MINSAL (including DIPOL, DIPRECE, EPI, DIPLAS, DIVAP, DIGERA, DIGEDEP, Infrastructure, among others) to face extreme high temperature events and heat waves, ensuring an effective and timely response at the national and regional level.													
	Indicator Measurement		1. Internal plan of the Ministry of Health for the response to extreme high temperatures and heat waves made, approved and disseminated.													
	Institutions		Responsible	Climate Change Unit, DEGREYD Central level: - SSP: DIPOL, EPI, DIPLAS, DIPRECE, others - MRS: DIGERA, DIVAP, DIGEDEP, others - Public Health Institute (ISP) Regional level: - Regional Ministerial Health Secretariats - Health Services Other Departments or Offices to which the plan assigns tasks or responsibilities												
	Reach		Beneficiaries	Reduction of the health effects on the population exposed to extreme high temperatures and heat waves												
			Territorial	National level												
Outcomes	Incorporates or contributes to the guidelines cross-sections of ECLP, PNACC, NDC		ENS: LT4													
	Synergy with other public policies		- Synergy with the SDGs: 3 on Good Health and Well-being, 13 on Climate Action - Gender responsive, since a Plan of this nature considers in its diagnosis the different IBBs that impact the groups of special attention. Intra-sectoral coordination (different levels of the health sector) to address climate change													
Planning measure	Timeline Implementation		Actions (Specify the name of the shares)	Years					Indicators Actions (quantitative or qualitative)					Institutions		Total cost per action
	(Main activities that			Year 1 2025	Year 2 2026	Year 3 2027	Year 4 2028	Year 5 2029	Name	Meta	United d the measure a	Medium of Verification n	Categorization n Media of Implementation on (T, C, F u O)19	Answers	Contributor ras	

¹⁹ Indicate if the action can be categorized within any of the following means:

T: Technology development and transfer.

C: Capacity building and strengthening

F: Financial guidelines.

O: Other.

must be done for the fulfillment of the goal of (Measurement)	Action 1 Convene a working group and define the scope of the plan.	X					Exempt resolution creating a working group for the plan	100	%	Office with Resolution n Exempt that creates a table job	C: Creation of capabilities	DEGREYD	Sector Private of Health, Societies Scientific, Municipalities, Corporation includes Municipal Health	
	Cost	MS 0 MS 0		MS 0	MS 0	MS 0								MS 0
	Action 2 Perform a diagnosis of the different instruments of the MINSAL to address events of extreme high temperatures and heat waves.		X				Gantt chart of the work	100	%	Paper Gantt	C: Capacity Building	DEGREYD	Sector Private of Health, Societies Scientific, Municipalities, Corporation includes Municipal Health	
	Cost	MS 0	MS 3.159	MS 0	MS 0	MS 0								MS 3.159
	Action 3 Identify gaps and develop the missing instruments to address the response to high-level events extreme temperatures and heat waves.			X			Reports from the working group (frequency to be defined by the group)	100	%	Working group report	C: Capacity Building	DEGREYD	Sector Private of Health, Societies Scientific, Municipalities, Corporation includes Municipal Health	
	Cost	MS 0 MS 0		MS 6.138 MS 0		MS 0								MS 6.138
	Action 4 Generate a proposed internal plan for Ministry of Health for the response to extreme high temperatures and heat waves, considering vulnerable groups requiring special attention.				X		MINSAL Plan developed	100	%	Plan Internal of the Ministry	C: Creation of capabilities	DEGREYD		
	Cost	MS 0	MS 0	MS 0	MS 15.796	MS 0								MS 15.796
	Action 5 Disseminate the plan to the local levels of health of the entire country.				X		Official communication informing the local levels of the plan	100	%		C: Creation of capabilities	DEGREYD		
	Cost	MS 0 MS 0		MS 0	MS 0	MS 0								MS 0
Total cost per year MS 0			MS 3.159	MS 6.139	MS 15.796	MS								MS 25.274
Financing	Estimate of cost of the measure	M\$ 25.274												
	Possible sources of funding	To be defined by the Ministry of Health. Additional budget is required.												
Information additional	Assumptions	The availability of DEGREYD officials for work on the tables, reports, etc. will be taken into account.												

Measures related to water security

One of the main impacts of climate change is the decrease in water resources and changes in their composition. Both impacts pose health risks, whether due to insufficient quantities of drinking water for domestic consumption or to chemical and microbiological parameters exceeding limits considered acceptable health risks.

The recognition of both impacts demands an adaptation of the monitoring and control in sources of collection and bodies that have sanitary use, mainly the quantity and quality of drinking water that the services distribute; and the quality of waters for recreation with direct contact.

Consequently, it has been considered necessary to prioritize a substantial improvement in the planning, reporting, and recording of health surveillance activities in services

of drinking water and strengthening the quality control of water used for recreation involving direct contact. Additionally, the focus and prioritization of activities allows for the efficient use of resources; therefore, social, health, and environmental variables have been incorporated into the diagnoses of the Regional Health Authorities (SEREMIS).

4.13.- Adaptation Measure No. 13: Sanitary control of emerging contaminants in freshwater bodies

Controlling the quality of water for recreation and human consumption requires adapting to new contaminants that increase health risks.

In this sense, this measure proposes to include a new activity not considered in the Climate Change Adaptation Plan for the period 2016-2022, to prepare and provide the health sector with the knowledge and tools that allow for a timely analysis of the main emerging pollutants that put people's health at risk, such as the presence of cyanotoxins.

Measure No. 13: "Inventory and Sanitary Control of Emerging Contaminants in Freshwater Bodies with Emphasis on Cyanotoxins"			
Element	Sub-element	Content	
Measurement	Aim	Strengthen water security to guarantee access to safe and clean water.	
	Specific (OE)		
	Lineament	E4 Climate risk reduction	
	Strategic (LE)		
	Description and expected result of the measure	One of the impacts of climate change that has been evidenced is the presence of emerging contaminants (e.g., cyanotoxins) in bodies of water that put the health of those who use them at risk, mainly for domestic human consumption and recreation with direct contact. As described in the literature, cyanotoxins are a type of contaminant whose presence and proliferation in continental surface water bodies is influenced by climate change. Although an inventory of the various contaminants in water bodies should be carried out, special emphasis will be placed on this agent. META: 1. Identify and characterize water bodies for recreational use and human consumption, currently impacted and susceptible to being impacted by emerging contaminants associated with the consequences of climate change. 2. Prioritize bodies of water for recreational use and human consumption with a medium or high risk of cyanotoxin presence that require analytical determination to specify the health risks associated with their use. 3. Implement analytical methodologies for the determination of cyanotoxins in prioritized regions. 4. Implement measures to prevent the health effects on people caused by exposure to bodies of water contaminated with cyanotoxins.	
	Indicator	1. Consolidated report with the identified and characterized bodies of water for recreational use and human consumption, currently impacted and susceptible to being impacted by emerging pollutants associated with the consequences of climate change prepared /Scheduled consolidated reports. 2. Consolidated report with the bodies of water for recreational use and human consumption with medium or high risk of presence of cyanotoxins that require an analytical determination to specify the health risks associated with their use prepared and published/committed consolidated reports. 3. Analytical methodologies for the determination of cyanotoxins in prioritized regions implemented developed/ Methodologies committed. 4. Guidance document with measures for the prevention of health effects on people caused by exposure to bodies of water contaminated with cyanotoxins prepared / Scheduled guidance documents.	
	Institutions	Head of the Environmental Health Department at DIPOL.	
Synergies	Reach	Beneficiaries	National population that uses water susceptible to being contaminated by the impacts of climate change, whether for recreational use or for human consumption.
			Identify contaminants in inland water bodies intended for human use and consumption
		Territorial	National
Measures	Incorporates or contributes to	ECLP	
	Cutting guidelines, PNACC, NDC	It can be linked to the water security aspect of other adaptation plans.	

	Synergy with other public policies	Synergy with the Sustainable Development Goals. Water security. Level of gender mainstreaming: Gender sensitive.													
	Timeline Implementation	Actions (Specify name of the actions)	Years					Indicators Actions (quantitative or qualitative)					Institutions		Total cost per action
			Year 1 2025	Year 2 2026	Year 3 2027	Year 4 2028	Year 5 2029	Name	Meta	Unit of measure	Means of Verification	Categorization Media of Implementation (T, C, F u O)20	Responsible for	Collaborators	
		Action 1 List the main continental bodies of water regional level that are used for recreational use and human consumption, and that present conditions of sanitary quality risk for recreational use and/or human consumption.	x					1.1.- Ordinary sent to SEREMIS Health 1.2.- Application from MINSAL to the DGA 1.3.- Consolidate the National	1.1.- Ordinary sent to SEREMIS Health 1.2.- Application for MINSAL to the DGA 1.3.- Consolidate do National	Number of Document you officers	1.1.- Official request (letter), from Central Level to the SEREMIS of the country's Health Department, information on freshwater bodies that present sanitary quality conditions that pose a risk to recreational use and human consumption in the region (define prioritization criteria: eutrophication, exposure, tendency to decrease, others). 1.2.- Official request for Information to the DGA on the characteristics of water bodies and their quality to evaluate options for joint and continuous collaboration that contribute to this measure. 1.3.- Document with the A national consolidated list containing the main continental water bodies at the regional level that are used for recreational use and human consumption, and that present sanitary quality conditions of risk for recreational use and human consumption in each region.	Other	Department the of Health Environmental of MINSAL	SEREMIS of Health ISP Deputy Directorate of Services Restrooms Rural MOP-DGA	MS\$36.251
Planning	(Main activities that must be carried out to achieve the goal of the Measure)	Cost	MS\$36.251	MS\$0	MS\$0	MS\$0									
		Action 2 Analyze and categorize the bodies of water characterized in the previous action to analytically determine the risk level of the pollutants, with emphasis on cyanotoxins	x					Report	Report	Number of Document the officer	Report with the results of the analysis to prioritize management or interventions in freshwater bodies for use and/or consumption.	Other	Department the of Health Environmental of MINSAL	Not applicable	MS\$3.522
		Cost	MS\$3.522	MS\$0	MS\$0	MS\$0									
		Action 3 Assess local capabilities and technical feasibility for implementing analysis of contaminants prioritized in localities specific, with emphasis on cyanotoxins, preferably close to the bodies of water for recreational use and human consumption, including sources of catchment.	x					3.1.- Job Seremis 3.2.- Report	3.1.- Job Seremis 3.2.- Report	Number of Document the officers	3.1.- Official letter to the SEREMIS of Health communicating the analysis carried out and prioritization of water bodies for Regional management. 3.2.- Report on the assessment of regional capacities at the national level	Other	3.1. DESAM 3.2.- Department or National and Reference in Health Environmental of ISP and the Office of Coordination of the Network Laboratories Health Public DIPOL	SEREMIS of Health	MS\$41.534
		Cost	MS\$41.534	MS\$0	MS\$0	MS\$0									
		Action 4 Develop a joint work plan ISP- DESAM- Of Coord Laboratories and	x					Plan of Job	Plan of Job	Number of Document the officer	Work Plan	Creation and strengthening to of capabilities	Department or Health Environmental, Department or National and Reference	Not Applicable	MS\$14.878

²⁰ Indicate if the action can be categorized within any of the following means:

T: Technology development and transfer.

C: Capacity building and strengthening

F: Financial guidelines.

O: Other.

[illegible]

4.14.- Adaptation Measure No. 14: Conduct a vulnerability assessment on territorial availability of water resources and sanitary conditions for its consumption

The increase in risk factors associated with the availability and quality of drinking water must be analyzed alongside other factors that exacerbate health impacts. Therefore, a measure has been incorporated that integrates health, social, and environmental information. This will allow the Regional Health Authorities (SEREMI) to determine priority areas for their actions and, in this way, reduce health risks for people, primarily in rural areas of the country most affected by the inability to collect water and make it potable for domestic consumption.

It is important to note that this measure is a continuation of the previous Climate Change Adaptation Plan of the Ministry of Health (Adaptation Measure No. 15, established under Action Axis No. 7), which involved developing a spatial representation of water vulnerabilities that impact the availability of water for human consumption. This Sectoral Adaptation Plan includes expanding the variables used for analysis to prioritize the activities that the Regional Health Authority (SEREMI) must carry out in the sanitary surveillance of drinking water.

Measure No. 14 "Conduct a vulnerability assessment on territorial availability of water resources and sanitary conditions for their consumption"		
Element	Sub-element	Content
of Description	Aim	Strengthen water security to guarantee access to safe and clean water.
	Specific (OE)	
	Lineament Strategic (LE)	E2 Analysis and Research on climate impacts and scenarios.

	Description and expected result of the measure	The dynamics of water scarcity and drought impact the availability of water for human consumption, which can worsen if the sanitary and social conditions of families are insufficient.													
		Given the above, it has been considered appropriate to develop a vulnerability diagnosis that considers water, health and social variables, in order to have information on rural areas of the regions that should be observed and used by the SEREMI of Health to avoid risks to human health derived from the shortage of drinking water-META: 1.- Gathering updated information on the state of water supplies in rural areas with unavailability of water for human consumption or with quality problems. 2.- Implement a vulnerability analysis methodology with water, health and social variables. 3.- Apply and disseminate with the SEREMI of Health the result of the methodology with the spatial representation of vulnerable areas.													
	Indicator Measurement	1. Report with updated information on the state of water supplies in rural areas with water unavailability for human consumption or with quality problems prepared / Reports with the collection of updated information on the state of water supplies in compromised rural areas. 2. Vulnerability analysis methodology with water, health and social variables developed and disseminated. 3. Report on the results of applying the methodology with the spatial representation of vulnerable areas disseminated among the Regional Health Authorities (SEREMIS) / Compromised reports.													
	Institutions	Responsible	DIPOL Environmental Health Department.												
	Reach	Beneficiaries	National population, predominantly rural, that consumes drinking water.												
Territorial		Focus water health surveillance activities on those supplies identified as vulnerable. National													
Synergies	It incorporates or contributes to the cross-cutting guidelines of ECLP, PNACC, NDC	ECLP ODS Water security of the social pillar of the NDC													
	Synergy with other public policies	Alignment with other public policies It can be linked to the water security aspect of other adaptation plans. Level of gender mainstreaming: Gender Sensitive													
Planning	Timeline Implementation (Main activities that must be carried out) for the fulfillment of the goal of (Measurement)	Actions (Specify the name of the shares)	Years					Indicators Actions (quantitative or qualitative)					Institutions		Total cost per action
			Year 1 2025	Year 2 2026	Year 3 2027	Year 4 2028	Year 5 2029	Name	Meta	Unit of measure a	Medium of Verification	Categorization Media of Implementation (T, C, F u O)21	Responsible	Contributors	
		Action 1 Prepare an information request and send it to the Regional Health Authority	x					1.1- Ordinary from MINSAL to Regional Health Secretariats 1.2.- Word. from MINSAL to DGA	1.1- Ordinary from MINSAL to Regional Health Secretariats 1.2.- Word. from MINSAL to DGA	Number the of Docu official statement	1.1.- Ordinary sent to the SEREMI of Health. 1.2.- Official call from MINSAL to the DGA to analyze collaboration options joint	Other	Department of Health Environmental	DGA - MOP	MS\$19.009
		Cost	MS\$19.009	MS \$ 0 MS 0		MS 0 MS 0									
		Action 2 Define and apply a methodology of health, water and social dimensions and variables to develop a vulnerability diagnosis.	x					Report	Report	Number the of Document Official	Report describing the methodology used and results of vulnerability obtained.	Creation and strengthening to develop capabilities	Department of Health Environmental	DEGREYD Deputy Directorate of Services Restrooms Rural MOP(DOH-DGOP-SOP)	MS\$10.430
		Cost	MS\$10.430	MS \$ 0 MS 0	MS 0 MS 0										
		Action 3 Order sending that reports the results, guides activities and calls for a meeting with regional teams from the drinking water programs.	x					Ordinary	Ordinary	Number the of Document Official	Ordinary with results report and guidelines sent; program and attendance list of Meeting with drinking water program teams from the Regional Health Authorities (SEREMIS) across the country. This will be copied to SENAPRED	Other	Department of Health Environmental	DEGREYD SENAPRED	MS\$1.761
		Cost	MS\$1.761	MS \$ 0 MS 0	MS 0 MS 0										
		Total cost per year	MS\$31.200	MS \$ 0 MS 0	MS 0 MS 0										MS\$31.200

²¹ Indicate if the action can be categorized within any of the following means:

T: Technology development and transfer.

C: Capacity building and strengthening

F: Financial guidelines.

O: Other.

Financing	Estimated cost of the measure (M\$)	M\$31.200
	Possible sources of funding	To be defined by the Ministry of Health. Additional budget is required.
Additional Information	Assumptions	- A license is obtained for the use of the methodology for weighting variables by the SEREMI of Health. - The regional teams generate the information necessary for the proper use of the methodology. - There is time available in the SEREMI of Health for the use of the methodology.

4.15.- Adaptation Measure No. 15: Modernize the sanitary surveillance of drinking water through digital applications with a risk-based approach

In the context of the impact of climate change on water resources, the health surveillance of drinking water carried out by the Regional Health Authorities (SEREMI de Salud) needs to be modernized to manage identified health risks in a timely manner. To this end, it is essential to design and implement digital systems that allow for the registration of drinking water supply services, the scheduling and reporting of water safety surveillance activities, and the reporting of the results of activities associated with a health risk.

Measure No. 15: “Modernize the sanitary surveillance of drinking water through digital applications with a risk-based approach.”						
Element	Sub-element	Content				
Measurement	Aim Specific (OE)	Strengthen water security to guarantee access to safe and clean water.				
	Lineament Strategic (LE)	E2 Analysis and Research on climate impacts and scenarios.				
	Description and result expected from the measure	The decrease in water resources impacts the availability of water that can be collected for human consumption, which in turn impacts the health of families that do not have sufficient and continuous access to drinking water or whose water quality has changed. Given the above, it has been considered to strengthen the sanitary surveillance of water for human consumption through the development of a digital application that allows, mainly the Regional Ministerial Secretariats of Health, to register water services, the sanitary surveillance activities they carry out in them and identify sanitary risk factors associated with the provision, prioritizing the sanitary services or provisions that operate in rural areas. META: 1.- Design and develop a health surveillance tool for drinking water on the MIDAS platform of the Ministry of Health, which includes the registration of supplies, results of activities (taking samples for laboratory analysis, inspections, field measurements) carried out on them, and weighting of health risk. 2.- Train the SEREMI of Health and monitor the use of the developed module. 3.- To implement and formalize the use of the module for prioritizing and registering the health surveillance activities of drinking water in the SEREMI of Health.				
	Indicator Measurement	1. Digital module for sanitary surveillance of drinking water developed on the MIDAS platform / Number of Surveillance Modules water committed				
	Institutions	Responsible Institution in charge of monitoring compliance with the adaptation measure (must be a single institution)				
	Reach	Beneficiaries	Population, predominantly rural, that may present a health risk in the provision of drinking water that supplies it. It also benefits people who perform functions in water inspection and health surveillance in regional health authorities, prioritizing activities in drinking water services with health risks. Optimize water health surveillance tasks by focusing on those supplies that present the greatest health risk			
	Territorial	National				
Synergies	Incorporates or contributes to the guidelines transversals by ECLP, PNACC, NDC	ECLP WE ODS Water security according to the Social Pillar Criterion of the NDC				
	Synergy with other public policies	Alignment with other public policies. It can be linked to the water security aspect of other adaptation plans. Level of gender mainstreaming: Gender Sensitive				
Plan to monitor the implementation	Actions (Specify)	Years	Indicators Actions (quantitative or qualitative)		Institutions	Total cost per action

		name of shares)	Year 1 2025	Year 2 2026	Year 3 2027	Year 4 2028	Year 5 2029	Name	Meta	Unit d of measure a	Medium of Verification	Categorization Media Implementation (T, C, F u O)22	Responsible for	Collaborators	
Timeline Implementation n (Main activities that must be carried out to achieve the goal of the Measure)		Action 1 Determine the procedure and identify the main actions and information to determine risk factors and their weighting	x (2024)					Module Surveillance of Water in MIDAS	Implementation tado	Number the of digital you	Health surveillance module of water for consumption human developed in platform MIDAS	Creation and strengthening of capabilities	Department or Health Environmental	- Office of Coordination of Red of Laboratories of Public health environmental and labor-related; and - Equipment Development MIDAS - TIC	MS77.994
		Cost	MS77. 994	MS 0	MS 0	MS 0	MS 0								
		Action 2 Prepare and conduct in-person and virtual training sessions for officials who, in the SEREMI of Health, perform functions in water sanitary surveillance.	x					Trigger capabilities in the use of surveillance module	Activity its from train me bring	Number the of Activid ades of They enter Amen to	Training report is carried out a officials and officials who perform surveillance functions health of waters in the SEREMI of Health	Creation and strengthening of capabilities Technology development and transfer	Department or Health Environmental	- Office of Coordination of Red of Laboratories of Public health environmental and labor-related; and - Equipment Development MIDAS - TIC	MS23.805
		Cost	MS23. 805	MS 0	MS 0	MS 0	MS 0								
		Action 3 To implement the pilot program and officially communicate to the SEREMI of Health development and availability of the module water sanitation surveillance in MIDAS.	x					Official use surveillance module	Ordinary processed	Number the of Docu ment official	Communication document official of the availability d of module for use by the SEREMIs Health.	Technology development and transfer Other	Department or Health Environmental	- Office of Coordination of Red of Laboratories of Public health environmental and labor-related; and - Equipment Development MIDAS - TIC	MS1.057
		Cost	MS10. 57	MS 0	MS 0	MS 0	MS 0								
		Action 4 To monitor and support the use of the module water health surveillance in the SEREMI of Health.	x	x	x	x	x	Report results of use module	Report Final	Docu ment official	Number of Report on results of the use of module in the SEREMI of Health.	Other	Department or Health Environmental	- Office of Coordination of Red of Laboratories of Public health environmental and labor-related; and - Equipment Development MIDAS - TIC	MS52.830
		Cost	MS10. 566	MS10. 566	MS10. 566	MS10. 566	MS10. 566								
		Action 5 Preparation and submission of an official letter to the DGA to analyze collaboration options	x					Order from MINSAL addressed to the DGA	Ordinary	Docu official statement	3.3.1.- Call for applications official since MINSAL to the DGA to analyze joint collaboration options	Other	Department or Health Environmental	DGA Deputy Directorate of Services Restrooms Rural	MS1.761
		Cost	\$1.76 1	MS 0	MS 0	MS 0	MS 0								
		Total cost per year	MS92. 290	MS10. 566	MS10. 566	MS10. 566	MS10. 566								MS157.447
Financing	Estimate of cost of the measure (M\$)	\$157.447													
	Possible sources of financing	To be defined by the Ministry of Health. Additional budget is required.													
Additional Information	Assumptions	- The necessary resources are available for the design and development of a digital module for water health surveillance. - The SEREMI de Salud has resources available to train and implement the digital module. - The resources are available to conduct effective training. - The module will be used permanently by the Regional Health Authorities (SEREMI).													

²² Indicate if the action can be categorized within any of the following means:

T: Technology development and transfer.

C: Capacity building and strengthening

F: Financial guidelines.

O: Other.

5.- Institutional coordination and arrangements for the implementation of the measures and the plan

To effectively implement the climate change plan in the health sector, each fact sheet specifies the responsibilities of the various departments involved and the coordination mechanisms. Furthermore, it describes the enabling actions for implementing the different sectoral measures, in accordance with the national strategies related to the means of implementation established in the Framework Law on Climate Change.

In this context, the Executive Coordination Unit for Climate Change, established by Resolution (E) No. 1290 of September 10, 2018, from the Minister of Health, is responsible for monitoring the plan's progress, given that the actions outlined in this plan fall under the responsibility of the various departments within the Ministry. To achieve its objective, monthly working group meetings will be held with the different stakeholders. The Ministerial representative to the ETICC is in charge of this Unit, thus ensuring ongoing communication with the Ministry of the Environment on these matters.

This plan is correlated with the mitigation plan presented by this ministry, as a link can be identified between measure No. 5 of the mitigation plan and measure No. 11 of this plan. On the one hand, the mitigation plan establishes energy efficiency and a reduction in the carbon footprint, and on the other hand, the adaptation plan establishes increased resilience of the facilities by ensuring the infrastructure for supplying vital lines for their operation.

6. Monitoring and Evaluation

Regarding the monitoring of this plan, the sectoral authority will report annually to the Ministry of the Environment on the progress of the sectoral plan and its measures. This activity must be repeated every year that the plan is in effect.

The objective of the monitoring will be to establish a hierarchical weighting system for the milestones, allowing the evaluation of the relative importance of each milestone within the measure and prioritizing resources and efforts according to their relevance to the fulfillment of the plan's objectives.

Each of the sheets described above contains monitoring tools, indices or indicators and control which will be broken down and analyzed with the AHP method which allows solving complex problems of multiple criteria by building a hierarchical model and organizing the information in an efficient and graphic way in the face of a given problem.

Monitoring the Sectoral Plan is crucial to ensuring the implementation of actions and learning from experiences. During the five-year term of the National Adaptation Plan, annual monitoring reports will be prepared for each Sectoral Plan, along with an annual report for the Plan itself, which will include a summary of sectoral progress, as shown in the example in the table.

Specific Objectives, Strategic Lines and Measures			Year 1			
AME TON	THE	Measures	Meta	Indicator	Means of Verification	Amount invested
O1. Promote healthy and sustainable eating	E1. Health education, training, awareness and cooperation.	Promoting healthy and sustainable eating.	Spread the word Feeding Healthy	Spread the word Feeding Healthy.	Annual report of dissemination activities carried out in the context of the implementation of the Dietary Guidelines, within the framework of the Plan Climate Change Adaptation.	M\$166.227
	E2. Analysis of e research on impacts and scenarios	Analysis of the dynamics between climate change and foodborne illnesses.	Circular	Official document	Updated circular	M\$134.745
	E3. Coordination intersectoral and interinstitutional	Prevention and Management poisonings caused by harmful algal bloom (HAB) events in marine waters.	Updated protocol	Document	Alert protocol early updated, according to the type of algal bloom in marine waters.	M\$ 1.712
		Prevention and Management poisonings caused by harmful algal bloom (HAB) events in marine waters.	Coordination between MINSAL – ISP and SEREMIS of Health to project biological supplies for the year	Document	Document containing the annual projection of biological inputs	M\$ 3.400
Barriers						
Assessment						

Regarding the evaluation, a monthly assessment will be conducted, based on the progress made, to analyze facilitators and barriers and propose improvement strategies. The Ministry of Health will be responsible for evaluating and updating the plan, and specifically, the reduction of vulnerability and the increase in resilience of healthcare facilities and staff will be considered. The results of these assessments will be used to update the plans for the next period.



Example of a consolidated report of measures according to the evaluation by the AHP method

7. Recommendations for the Territorial Scale

The document presents several specific recommendations for the country's regions in the context of climate change adaptation. These recommendations include:

- Regional emergency plans implement an existing and widely disseminated methodology that allows their development to consider both locally relevant climate threats and local non-climate threats, using updated risk matrices to ensure adequate preparedness and response.
- Vulnerability assessment and preparedness for climate events: It is suggested that assessments be conducted on territorial water availability and the different sanitary conditions for its consumption according to the area, focusing on rural areas where access to drinking water is limited. The assessment should utilize existing technological tools (MIDAS). Implementing a vulnerability analysis methodology that includes water, sanitation, and social variables to focus the actions of the Regional Health Authorities (SEREMI).
- Conducting a survey and subsequent implementation of the lifelines of each high and medium complexity establishment will allow the establishments to build resilience to climate change.
- Control of cyanotoxins in water bodies: It is proposed to prioritize the analysis and control of emerging contaminants, such as cyanotoxins, in freshwater bodies used for recreation and human consumption. This requires establishing sampling protocols and analytical methodologies in prioritized regions.
- Training and institutional strengthening: It is essential to develop the capacities of the Regional Health Authorities (SEREMI) and other regional entities to improve water health surveillance and the response to climate risks, through training and the implementation of monitoring systems. This also includes promoting healthy eating, which may vary depending on the area or region where the recommendations are given.
- Climate Risk Reduction: Epidemiological surveillance of the health effects associated with climate change is important at the local level since the analysis of records of communicable and non-communicable diseases will allow for anticipation of events that may occur.

Furthermore, the healthcare network in the different areas must take appropriate measures regarding the response to heat waves as well as to the appearance of vectors and reservoirs.

These measures seek to strengthen the resilience of regions to climate risks, focusing on water resource management, pollutant control, and institutional capacity for effective response.

8. Institutional, regulatory and economic barriers

Regarding possible barriers to compliance with the measures and means of implementation of the Plan, it should be noted, in the institutional sphere, the structure of the health sector, in which a significant part of the organizations that comprise it are functionally decentralized bodies, endowed with legal personality and their own assets, and enjoy a high degree

of autonomy, as is the case of the Health Services and the Public Health Institute, all of which will have to play an important role in the future implementation of the Sectoral Adaptation Plan.

To overcome this barrier, the role of the central sectoral authority will be fundamental, which, in its capacity as responsible for national health policy, exercises a directive and guiding function, both during the implementation of the Sectoral Adaptation Plan itself, and during the formulation and implementation of the regional and communal Climate Action Plans.

Regarding regulatory barriers, it should be noted that, in the area of environmental health—a key area where sectoral adaptation measures may require regulatory support—the health sector shares regulatory powers with the Ministry of the Environment, although differences in their regulatory priorities may exist. Overcoming any such barriers will depend on the level of coordination and collaboration achieved between the two ministries.

Regarding economic barriers, the allocation of sectoral resources necessary for the implementation of the plan may be severely disrupted by unforeseen changes in the global and/or national epidemiological situation that require a redefinition of priorities and resources, as was the case with the recent Covid-19 pandemic, or as could occur in the face of the eventual emergence of relevant epidemiological events associated with emerging and re-emerging diseases.

9. Prioritization of measures

Prioritizing measures is a significant challenge, as cost-benefit analysis reveals that the measures entail high costs compared to their potential benefits. This assessment is further complicated by the fact that the benefits are estimates that may vary depending on the actual effects in each country.

The corresponding analysis will be carried out in order to prioritize measures according to the resources that are allocated, an analysis that will be carried out using various approaches of relevance in health, instead of adopting a cost-benefit approach.

- Risk Analysis:
Prioritize measures based on risk assessment, identifying those actions that address the most critical and urgent risks to public health.
- Impact on Health:
Evaluate and prioritize measures according to their direct impact on the health of the population, giving priority to those that have the potential to prevent serious diseases or significantly improve public health.
- Technical and Operational Feasibility:
Consider the technical and operational feasibility of each measure, prioritizing those that are easiest to implement and maintain with the available resources and capabilities.
- Equity and Social Justice:
Prioritize measures that promote equity and social justice, ensuring that actions benefit the most vulnerable and disadvantaged populations.

- Political and Social Acceptability:

Evaluate the political and social acceptability of the measures, prioritizing those that have the greatest support from stakeholders and the community.

- Synergies and Co-benefits:

Identify measures that generate co-benefits, that is, that in addition to addressing climate change, also contribute to other sustainable development goals, such as poverty reduction or improved air quality.

Each of these approaches can provide a different and valuable perspective for prioritizing measures, and in many cases, a combination of several approaches may be the most effective strategy.

10. Schedule and costs of the measures

The following table provides an overview of the implementation timelines for the various adaptation measures. It also identifies the responsible institution and relevant stakeholders involved in the execution of each measure.

Measures	Actions	Years					Total cost per action
		2025	2026	2027	2028	2029	
Measure No. 1: "Promotion of healthy and sustainable eating"	Action 1	x	x	x	x	x	M\$669,633
Measure No. 2: "Analysis of the dynamics between climate change and foodborne illnesses"	Action 1		x				M\$35,912
	Action 2:	x	x	x	x	x	M\$480,745
Measure No. 3: "Prevention and management of poisonings caused by harmful algal bloom (HAB) events in marine waters."	Action 1	x					M\$ 1,712
	Action 2:		x				M\$ 2,000
	Action 3:		x	x	x	x	M\$3,979,004
	Action 4	x	x	x	x	x	M\$19,200
Measure No. 4: "Monitoring and communication system for unusual findings, in time and space, of vectors and reservoirs of public health importance prioritized* that are associated with climate change."	Action 1	x					M\$8,745
	Action 2:	x					M\$98,364
	Action 3:		x	x	x	x	M\$90,792
Measure No. 5: "Strengthening the response capacity of the healthcare network to vectors and reservoirs."	Action 1	x	x				M\$ 0
	Action 2:			x			M\$ 0
Measure No. 6: Strengthening public health human capital	Action 1	x					M\$ 11,507
	Action 2:	x	x	x	x	x	M\$ 20,938
	Action 3:	x	x	x	x	x	M\$ 18,760
Adaptation Measure No. 7: "Epidemiological surveillance system for health effects associated with climate change"	Action 1	x		x		x	M\$56,865
	Action 2:	x		x		x	M\$6,000
Measure No. 8: "Increasing the number of online health services available to avoid exposure to climate risks."	Action 1	x					M\$ 0
	Action 2:		x	x			M\$ 0
	Action 3:				x		M\$ 0
Measure No. 9: "Development of a response plan for the healthcare network to heat waves."	Action 1						M\$ 0
	Action 2:						M\$ 0
	Action 1	x					M\$ 18,293

Measure No. 10: "Implement a methodology that allows the development of regional emergency plans considering climate threats along with other local threats."	Action 2:		x	x	x	x	M\$ 0
Measure No. 11: A. "Diagnosis of vital lines in health establishments to face extreme weather events"	Action 1	x					M\$ 0
	Action 2:	x					M\$ 0
	Action 3:		x				M\$24.000
	Action 4			x			M\$24.000
	Action 5:				x		M\$24.000
	Action 6:				x		M\$24.000
Measure No. 11: B. "Implementation of improvements to lifelines in health facilities to cope with extreme weather events"	Action 1			x	x		M\$ 48.000
	Action 2:			x	x		M\$10.000.000
	Action 3:				x	x	M\$48.000
	Action 4					x	M\$5.000.000
Measure No. 12: "Development of an internal plan by the Ministry of Health for responding to extreme high temperatures and heat waves."	Action 1	x					M\$ 0
	Action 2:		x				M\$ 3.159
	Action 3:			x			M\$ 6.138
	Action 4				x		M\$ 15.796
	Action 5:				x		M\$ 0
Measure No. 13: "Inventory and Sanitary Control of Emerging Contaminants in Freshwater Bodies with Emphasis on Cyanotoxins"	Action 1	x					M\$36.251
	Action 2:	x					M\$3.522
	Action 3:	x					M\$41.534
	Action 4	x					M\$14.878
	Action 5:		x				M\$25.153
	Action 6:		x	x	x	x	M\$1.544.072
	Action 7		x	x	x	x	M\$238.900
	Action 8:		x	x	x	x	M\$1.985
	Action 9:	x		x			M\$10.567
	Action 10			x			M\$10.566
Measure No. 14 "Conduct a vulnerability assessment regarding territorial availability of water resources and sanitary conditions for their consumption"	Action 1	x					M\$19.009
	Action 2:	x					M\$10.430
	Action 3:	x					M\$1.761
Measure No. 15: "Modernize the sanitary surveillance of drinking water through digital applications with a risk-based approach."	Action 1	x					M\$77.994
	Action 2:	x					M\$23.805
	Action 3:	x					M\$1.057
	Action 4	x	x	x	x	x	M\$52.830
	Action 5:	x					M\$1.761

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